Book of Proof: Condensed Version with All Exercise Solutions

KS

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Part I
Fundamentals of Set Theory

1 Introduction to Sets

All of mathematics can be described with sets. Set theory is *a* perfectly suitable language to express and describe all types of mathematical structures.

Think of a *set* as a collection of things. The things are called *elements* of the set. It's somewhat worth noting here that we're only concerned with sets that express mathematical entities like functions, numbers, points, etc. Set construction: $\{2, 3, 4, 5\}$ is a set which has 4 elements. Some sets have infinitely many elements, *not all* sets are finite.

We say that two sets are *equal* if they contain exactly the same elements. So: $\{2, 3, 4, 5\} = \{2, 4, 3, 5\}$ but $\{2, 3, 4, 5\} \neq \{2, 4, 6, 8\}$.

To express that an element is *member* of a set, we write $2 \in A$, where A is a set containing 2 as an element. Some sets are very important and we reserve special symbols for them, like:

$$\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}, \mathbb{C}$$

Sets don't have to contain only numbers, for example $B = \{T, F\}$ or $C = \{a, b, c, d, e\}$, or $D = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$. A set can have *other sets as elements*:

$$K = \{1, \{2, 3\}, \{2, 4\}\}$$

It's really important to note here that K has 3 elements. $1, \{2, 3\}, \{2, 4\} \in K$ but $2, 3, 4 \notin K$.

If a set X is finite, we say that X's *cardinality* is the number of elements it has, and this number is denoted by $|X|$. An example of a special set is $\varnothing$. It is called the empty set and it enjoys the unique status of having no members and thus $|\varnothing| = 0$.

We have special *set builder notation* to describe sets that are too large or too complex to list between braces. Consider the infinite set of even integers $E = \{\dots, -4, -2, 0, 2, 4, 6 \dots\}$. We can instead write

$$E = \{2n | n \in \mathbb{Z}\}.$$

We read it 'the set of all things of form' $2n$, | 'such that', n 'is a member of' the set of integers. The following generalised structure follows for set-builder notation:

$$X = \{\text{expression} : \text{rule}\};$$

where the elements of X are understood to be *all values of expression* that are specified by the *rule*.

Example 1.0.1.

1. $\{n : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17 \cdots\}$
2. $\{n \in \mathbb{N} : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17, \dots\}$
3. $\{x \in \mathbb{R} : x^2 - 2 = 0\} = \{\sqrt{2}, -\sqrt{2}\}$
4. $\{x \in \mathbb{Z} : x^2 - 2 = 0\} = \varnothing$
5. $\{x \in \mathbb{Z} : |2x| < 4\} = \{-1, 0, 1\}$

Example 1.0.2. Describe the set $A = \{7a + 3b : a, b \in \mathbb{Z}\}$.

Solution. The set A is equivalent to the expression $7a + 3b$, where the rule for finding a, b is that they are members of the set of integers. Consider which integers. Let n be any integer, then $n = 7n + 3(-2n)$, such that $n = a$ and $b = -2n$. So, $n \in A$. It follows that A contains only integers, and furthermore, *every integer is an element of* A. So $A = \mathbb{Z}$. □

Important: Set-builder notation for $\mathbb{Q}$:

$$\mathbb{Q} = \{x : x = \frac{m}{n} \text{ where } m, n \in \mathbb{Z} \wedge n \neq 0\}$$

1.1 Exercises from Section 1

Exercise 1.1. Write each of the following sets by listing their elements between braces.

$1.\{5x - 1 : x \in \mathbb{Z}\}$ $9.\{x \in \mathbb{R} : \sin \pi x = 0\}$
$2.\{3x + 2 : x \in \mathbb{Z}\}$ $10.\{x \in \mathbb{R} : \cos x = 1\}$
$3.\{x \in \mathbb{Z} : -2 \leq x \leq 7\}$ $11.\{x \in \mathbb{Z} : |x| < 5\}$
$4.\{x \in \mathbb{N} : -2 < x \leq 7\}$ $12.\{x \in \mathbb{Z} : |2x| < 5\}$
$5.\{x \in \mathbb{R} : x^2 = 3\}$ $13.\{x \in \mathbb{Z} : |6x| < 5\}$
$6.\{x \in \mathbb{R} : x^2 = 9\}$ $14.\{5x : x \in \mathbb{Z}, |2x| \leq 8\}$
$7.\{x \in \mathbb{R} : x^2 + 5x = -6\}$ $15.\{5a + 2b : a, b \in \mathbb{Z}\}$
$8.\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\}$ $16.\{6a + 2b : a, b \in \mathbb{Z}\}$.

[Go to Solution]

Exercise 1.2. Write each of the following sets in set-builder notation.

$17.\{2, 4, 8, 16, 32, 64, \dots\}$ $23.\{3, 4, 5, 6, 7, 8\}$
$18.\{0, 4, 16, 36, 64, 100, \dots\}$ $24.\{-4, -3, -2, -1, 0, 1, 2\}$
$19.\{\dots, -6, -3, 0, 3, 6, 9, 12, 15\}$ $25.\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\}$
$20.\{\dots, -8, -3, 2, 7, 12, 17, \dots\}$ $26.\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, \dots\}$
$21.\{0, 1, 4, 9, 16, 25, 36, \dots\}$ $27.\{\dots, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, \dots\}$
$22.\{3, 6, 11, 18, 27, 38, \dots\}$ $28.\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\}$.

[Go to Solution]

Exercise 1.3. Find the following cardinalities.

$29.|\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}|$ $34.|\{x \in \mathbb{N} : |x| < 10\}|$
$30.|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}|$ $35.|\{x \in \mathbb{Z} : x^2 < 10\}|$
$31.|\{\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}\}|$ $36.|\{x \in \mathbb{N} : x^2 < 10\}|$
$32.|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}\}|$ $37.|\{x \in \mathbb{N} : x^2 < 0\}|$
$33.|\{x \in \mathbb{Z} : |x| < 10\}|$ $38.|\{x \in \mathbb{N} : 5x \leq 20\}|$.

[Go to Solution]

Exercise 1.4. Sketch the following sets of points in the xy-plane.

$39.\{(x,y) : x \in [1,2], y \in [1,2]\}$
$40.\{(x,y) : x \in [0,1], y \in [1,2]\}$
$41.\{(x,y) : x \in [-1,1], y = 1\}$
$42.\{(x,y) : x = 2, y \in [0,1]\}$
$43.\{(x,y) : |x| = 2, y \in [0,1]\}$
$44.\{(x,x^2) : x \in \mathbb{R}\}$
$45.\{(x,y) : x,y \in \mathbb{R}, x^2+y^2 = 1\}$
$46.\{(x,y) : x,y \in \mathbb{R}, x^2+y^2 \leq 1\}$
$47.\{(x,y) : x,y \in \mathbb{R}, y \geq x^2-1\}$
$48.\{(x,y) : x,y \in \mathbb{R}, x > 1\}$
$49.\{(x,x+y) : x \in \mathbb{R}, y \in \mathbb{Z}\}$
$50.\{(x,\frac{x^2}{y}) : x \in \mathbb{R}, y \in \mathbb{N}\}$
$51.\{(x,y) \in \mathbb{R}^2 : (y-x)(y+x) = 0\}$
$52.\{(x,y) \in \mathbb{R}^2 : (y-x^2)(y+x^2) = 0\}$.

[Go to Solution]

2 The Cartesian Product

Think of the Cartesian product as a type of multiplication.

$$A \times B$$

In order to understand products better we will define an *ordered pair*.

Definition 2.0.1. An ordered pair is a list of the form x, y of two things, namely x and y, enclosed in parenthesis and separated by a comma.[1]

There are many ordered pairs, consider the ordered pair $(2,4)$. Now, if we swap the numbers around, that would be an ordered pair as well, i.e. $(4,2)$. However, these two pairs are not identical, meaning

$$(2,4) \neq (4,2).$$

Ordered pairs can be used to describe points on a plane (as it is done in Calculus and Linear Algebra for example). Similarly to our previous talk about sets, ordered pairs need not be *just* numbers, they can be letters as well.

Example 2.0.1. The pair $(\{2,3\},\{3,4\})$ can be an ordered pair of two sets, say A, B, then $(\{2,3\},\{3,4\}) = (A,B)$;
(α,β) can also be a pair; finally, $((2,3),(3,4))$ can be an ordered pair of an ordered pair.

So, the following definition of Cartesian Products is important:

Definition 2.0.2. The Cartesian Product of two sets A, B is a set, denoted by

$$A \times B$$

and defined as

$$A \times B = \{(a,b) : a \in A, b \in B\}.$$

[1] Additionally, one can use the Kuratowski ordered pair definition, i.e. $\{\{x\},\{x,y\}\} = (x,y)$. However, the one given above, even though slightly cryptic is easier to grasp for someone who has never encountered sets before.

The following is a more visual example of how ordered pairs are formed.

Example 2.0.2. Let $A = \{k, l, m\}, B = \{o, p\}$ for A, B some sets, then the product of A with B is

$$A \times B = \{(k, o), (k, p), (l, o), (l, p), (m, o), (m, p)\}.$$

Schematically, one can think of these pairs in terms of rows and columns or in terms of a corrdinate system where A is the x-axis and B is the y-axis.

The following is a fact:

Fact 2.0.1. If A, B are finite sets, then $|A \times B| = |A| \cdot |B|$.

Example 2.0.3. Let $A = \{1, 2, 3, 4, 5, 6\}$ and suppose these are dice values. The Cartesian Product of A with itself is

$$A \times A = A^2.$$

The product of A with itself will be a finite set of ordered pairs.

$$|A \times A| = |A| \cdot |A| = 6 \cdot 6 = 36.$$

Think of the product as rolling a dice twice an recording the outcome, where the first roll corresponds to the first result and the second to the second result.

Example 2.0.4. Another few very useful examples are the sets:

$$\begin{aligned} \mathbb{R} \times \mathbb{R} &= \{(x, y) : x, y \in \mathbb{R}\} \\ \mathbb{R} \times \mathbb{N} &= \{(x, y) : x \in \mathbb{R}, y \in \mathbb{N}\} \\ \mathbb{N}^2 &= \{(x, y) : x, y \in \mathbb{N}\}. \end{aligned}$$

The first example is best described as a coordinate system or an xy-plane. The second example is simply all points on the plane, whose second coordinate is a natural number. The third example, is best described as the first quadrant of the xy-plane, where every x, y are natural numbers. More formally, the last example is called the *lattice points* of the first quadrant.

It is possible for one factor of a Cartesian product to be a Cartesian Product itself. Suppose the following:

$$\mathbb{R} \times (\mathbb{N} \times \mathbb{Z}) = \{x, (y, z) : x \in \mathbb{R}, (y, z) \in \mathbb{N} \times \mathbb{Z}\}.$$

There is a different way to think about these pairs. They are called *ordered triples*. The brackets are optional in the above example, one can also write

$$\mathbb{R} \times \mathbb{N} \times Z = \{(x, y, z) : x \in \mathbb{R}, y \in \mathbb{N}, z \in Z\}.$$

In general we can define

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_1 \in A_1, x_2 \in A_2, \ldots, x_n \in A_n\},$$

or more succinctly,

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_i \in A_i\} \text{ for } i = \{1, 2, 3, \ldots, n\}$$

Example 2.0.5. Represent the two sides of a coin by the set $S = \{H, T\}$, the possible outcomes of tossing the coin seven times in a row can be described as:

Solution. This is the representation

$$S^7 = \{(x_1, x_2, \ldots, x_7) : x_1, x_2, \ldots, x_7 \in S\}.$$

Now to find the possible outcomes:

$$|S^7| = 2 \cdot 2 \cdot \ldots 7 = 128.$$

□

I believe the following to be clear but still worth noting. Notice that the cardinality of the product of S with itself seven times is written inside the || and not outside. This might seem like a silly thing to say, however syntax is very important.

2.1 Exercises from Section 2

Exercise 2.1. Write out the indicated sets by listing their elements between braces.

1. Suppose $A = \{1, 2, 3, 4\}, B = \{a, c\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A \times A$
   $(d) B \times B$
   $(e) \varnothing \times B$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) B^3$

2. Suppose $A = \{\pi, e, 0\}, B = \{0, 1\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A^2$
   $(d) B^2$
   $(e) A \times \varnothing$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) A \times B \times B.$

3. $\{x \in \mathbb{R} : x^2 = 2\} \times \{a, c, e\}$.

4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\}$.
5. $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$.
6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\}$.
7. $\{\varnothing\} \times \{0, \varnothing\} \times \{0, 1\}$.
8. $\{0, 1\}^4$.

[Go to Solution]

Exercise 2.2. Sketch these Cartesian Products on the xy-plane (or in $\mathbb{R}^3$ for the last two).

$(9)\{1, 2, 3\} \times \{-1, 0, 1\}$	$(15)\{1\} \times [0, 1]$
$(10)\{-1, 0, 1\} \times \{1, 2, 3\}$	$(16)[0, 1] \times \{1\}$
$(11)[0, 1] \times [0, 1]$	$(17)\mathbb{N} \times \mathbb{Z}$
$(12)[-1, 1] \times [1, 2]$	$(18)\mathbb{Z}^2$
$(13)\{1, 1.5, 2\} \times [1, 2]$	$(19)[0, 1]^3$
$(14)[1, 2] \times \{1, 1.5, 2\}$	$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0, 1]$

[Go to Solution]

3 Subsets

It is common that every element, or some elements of a set A are also elements of another set B. Consider the following:

$$A = \{1, 2, 3\}, B = \{0, 1, 2, 3, 4, 5\}.$$

In cases as this one, we say that $A \subseteq B$, we might even say that $A \subset B$ (since all elements of A are contained within B).

Definition 3.0.1. Suppose A, B are sets. If every element of A is also an element of B, then we say that $A \subseteq B$. We write $A \nsubseteq B$ if A is *not* a subset of B i.e., if it is not true that *every* element of A is also an element of B. So $A \nsubseteq B$ tells us that *at least one* element of the set on the left is not contained in the set on the right.

Example 3.0.1. Some key examples are the following:

1. $\{2, 3, 7\} \subseteq \{2, 3, 4, 5, 6, 7\}$;
2. $\{2, 3, 7\} \nsubseteq \{2, 3, 4, 5, 6\}$;
3. $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$;
4. $\mathbb{R} \times \mathbb{N} \subseteq \mathbb{R}^2$;
5. $A \subseteq A$, any set A is a subset of itself;
6. $\varnothing \subseteq \varnothing$.

These examples bring out an important fact about the empty set.

$$\boxed{\text{If } B \text{ is any set whatsoever, then } \varnothing \subseteq B}.$$

Fact 3.0.1. The empty set is a subset of all sets, that is, $\varnothing \subseteq B$ for any set B.

Proof. Let B be any set, then consider the definition of subset. Then if $\varnothing \not\subseteq B$ this would mean that there are *some* members of $\varnothing$ which are not members of B. However, we know that by definition $x \notin \varnothing$, or more formally

$$\varnothing = \{x : x \neq x\}.$$

Therefore, $\varnothing \subseteq B$. □

Comment: This proof technique is called *Proof by Contradiction* and it does come up a lot in the exercises later on, so it is worth trying to understand what is happening at this stage, however, the proof strategy will be explained in detail later.

Fact 3.0.2. If $|B| = n$, then B has 2^n subsets. Further, for *any finite* set B that has n number of elements, then B has 2^n subsets.

Example 3.0.2. Consider the subsets of the set $B = \{1, 2, \{1, 3\}\}$, there are 3 elements so 2^3.

$$\{\varnothing, \{1\}, \{2\}, \{\{1, 3\}\}, \{1, 2\}, \{1, 2, \{1, 3\}\}, \{1, \{1, 3\}\}, \{2, \{1, 3\}\}\}.$$

Important: It is really important to notice that $\{1, 3\}$ is **not** a subset of B. As in:

$$\{1, 3\} \not\subseteq B \text{ but } \{1, 3\} \in B, \text{ however, } \{\{1, 3\}\} \subseteq B.$$

3.1 Exercises from Section 3

Exercise 3.1. List all the subsets of the following sets:

1. $\{1, 2, 3, 4\}$
2. $\{1, 2, \varnothing\}$
3. $\{\{\mathbb{R}\}\}$
4. $\varnothing$
5. $\{\varnothing\}$
6. $\{\mathbb{R}, \mathbb{N}, \mathbb{Q}\}$
7. $\{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\}$
8. $\{\{0, 1\}, \{0, 1, \{2\}\}, \{0\}\}$

[Go to Solution]

Exercise 3.2. Write out the following sets by listing their elements between braces.

1. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 2\}$

2. $\{X \subseteq \mathbb{N} : |X| \leq 1\}$
3. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 4\}$
4. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 1\}$

[Go to Solution]

Exercise 3.3. Decide if the following statements are true or false. Explain.

1. $\mathbb{R}^3 \subseteq \mathbb{R}^3$
2. $\mathbb{R}^2 \subseteq \mathbb{R}^3$
3. $\{(x, y) \in \mathbb{R}^2 : x - 1 = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\}$
4. $\{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x - 1 = 0\}$

[Go to Solution]

4 Powersets

Definition 4.0.1. If A is a set, the *power set* of A is another set, denoted $\mathcal{P}(A)$ and defined to be the set of *all* subsets of A. Formally,

$$\mathcal{P}(A) = \{X : X \subseteq A\}.$$

Suppose that we have some set $A = \{1, 2, 3\}$, then

$$\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

A *finite* set of n-elements has 2^n subsets and the powerset of that set has 2^n-elements.

Fact 4.0.1. If A is a finite set, then $|\mathcal{P}(A)| = 2^{|A|}$.

Example 4.0.1.

$$\mathcal{P}(\{a\}) \times \mathcal{P}(\{\varnothing\}) = \{(\{a\}, \{\varnothing\}), (\{a\}, \varnothing), (\varnothing, \varnothing), (\varnothing, \{\varnothing\})\}.$$

This example might appear a little bit intimidating, however consider the following:

$$\mathcal{P}(\{a\}) = \{\varnothing, \{a\}\} \quad \mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}.$$

Then we simply pair the elements together.

Next, some *wrong* facts about powersets.

1. $\mathcal{P}(1) = \{\varnothing, \{1\}\}$ is meaningless, since 1 is not a set.[2]
2. $\mathcal{P}(\{1, \{1, 2\}\}) = \{\varnothing, \{1\}, \{\{1, 2\}\}, \{1, \{1, 2\}\}\}$ becuase $\{1, 2\} \not\subseteq \{1, \{1, 2\}\}$, however $\{1, 2\} \in \{\{1, \{1, 2\}\}\}$.[3]
3. $\mathcal{P}(\{1, \{1, 2\}\})$ but $\{\{1\}\} \not\subseteq \{1, \{1, 2\}\}$, the reasoing is similar as in the previous footnote.

It is not possible to list all the members of a powerset of A if A is *infinite.*

[2] There could be some debate here, this is what is says in the textbook, so I will follow it, however we can express $1 = \{\varnothing\}$ or $\{\varnothing, \{\varnothing\}\}$ when we consider the V-hierarchy of set construction.

[3] The correct statement would be $\{\{1, 2\}\} \in \mathcal{P}(\{1, \{1, 2\}\})$.

4.1 Exercises From Section 4

Exercise 4.1. Write the following sets by listing their elements between braces.

1. $\mathcal{P}(\{\{a,b\},\{c\}\})$
2. $\mathcal{P}(\{1,2,3,4\})$
3. $\mathcal{P}(\{\{\varnothing\},5\})$
4. $\mathcal{P}(\{\mathbb{R},\mathbb{Q}\})$
5. $\mathcal{P}(\mathcal{P}(\{2\}))$
6. $\mathcal{P}(\{1,2\})\times\mathcal{P}(\{3\})$
7. $\mathcal{P}(\{a,b\})\times\mathcal{P}(\{0,1\})$
8. $\mathcal{P}(\{1,2\}\times\{3\})$
9. $\mathcal{P}(\{a,b\}\times\{0\})$
10. $\{X\in\mathcal{P}(\{1,2,3\}):|X|\leq 1\}$
11. $\{X\subseteq\mathcal{P}(\{1,2,3\}):|X|\leq 1\}$
12. $\{X\in\mathcal{P}(\{1,2,3\}):2\in X\}$.

[Go to Solution]

Exercise 4.2. Suppose that $|A|=m,|B|=n$. Find the following cardinalities.

1. $|\mathcal{P}\mathcal{P}\mathcal{P}(A)|$
2. $|\mathcal{P}\mathcal{P}(A)|$
3. $|\mathcal{P}(A\times B)|$
4. $|\mathcal{P}(A)\times\mathcal{P}(B)|$
5. $|\{X\in\mathcal{P}(A):|X|\leq 1\}|$
6. $|\mathcal{P}(A\times\mathcal{P}(B))|$
7. $|\mathcal{P}\mathcal{P}\mathcal{P}(A\times\varnothing)|$
8. $|\{X\subseteq\mathcal{P}(A):|X|\leq 1\}|$.

[Go to Solution]

5 Union, Intersection, Difference

The same way numbers can be combined with operations such as addition, multiplication, subtraction, and division, we have similar operations for sets too.

Definition 5.0.1. Let A, B be sets then:

1. The *union* of A, B is $A \cup B = \{x : x \in A \vee x \in B\}$
2. The *intersection* of A, B is $A \cap B = \{x : x \in A \wedge x \in B\}$
3. The *difference* of A, B is $A - B = \{x : x \in A \wedge x \notin B\}$

Let us consider the following examples:

Example 5.0.1. Let $A = \{a, b, c, d, e\}, B = \{d, e, f\}, C = \{1, 2, 3\}$:

1. $A \cup B = \{a, b, c, d, e, f\}$
2. $A \cap B = \{d, e\}$
3. $A - B = \{a, b, c\}$
4. $B - A = \{f\}$
5. $(A - B) \cup (B - A) = \{a, b, c\} \cup \{f\} = \{a, b, c, f\}$
6. $A \cup C = \{a, b, c, d, e, 1, 2, 3\}$
7. $A \cap C = \varnothing$
8. $A - C = \{a, b, c, d, e\}$
9. $(A \cap C) \cup (A - C) = \varnothing \cup \{a, b, c, d, e\} = \{a, b, c, d, e\}$
10. $(A \cap B) \times B = \{d, e\} \times \{d, e, f\} = \{(d, d), (d, e), (d, f), (e, d), (e, e), (e, f)\}$
11. $(A \times C) \cap (B \times C) = \{(d, 1), (d, 2), (d, 3), (e, 1), (e, 2), (e, 3)\}$.

Important: It is always true that $X \cap Y = Y \cap X, X \cup Y = Y \cup X$. In other words, these are *commutatitve operations.* However, in general, $X - Y \neq Y - X$.

5.1 Exercises From Section 5

Exercise 5.1. Suppose $A = \{4, 3, 6, 7, 1, 9\}, B = \{5, 6, 8, 4\}, C = \{5, 8, 4\}$. Find:

1. $A \cup B$
2. $A \cap B$
3. $A - B$
4. $A - C$
5. $B - A$

6. $A \cap C$

7. $B \cap C$

8. $B \cup C$

9. $C - B$

[Go to Solution]

Exercise 5.2. Suppose $A = \{0, 2, 4, 6, 8\}, B = \{1, 3, 5, 7\}, C = \{2, 8, 4\}$. Use the operations from exercise 5.1[4]

[Go to Solution]

6 Complements

The notion of a complement requires the idea of an *universal set*. The way I think about it is thus. Let P be the set of prime numbers, then if asked to give a number outside of P, one can point out to examples of composite numbers such as $4, 6, \cdots$. So there is an underlying assumption that P is part of a bigger set which is the set of natural numbers, i.e. $P \subseteq \mathbb{N}$. In this case we call $\mathbb{N}$ the *universe* or the *universal set.*

Definition 6.0.1. Let A be a set with a universal set U. The complement of A, denoted $A^{\complement}$ is the set $A^{\complement} = U - A$.

So, if P is the set of prime numbers, then:

$$P^{\complement} = \mathbb{N} - P = \{1, 4, 6, 8, 9, 10 \dots\}.$$

[5] and so we can say that $P^{\complement}$ is the set of all *composite numbers and* 1.

6.1 Exercises for Section 6

Exercise 6.1. Let $A = \{4, 3, 6, 7, 1, 9\}, B = \{5, 6, 8, 4\}$ have universal set $U = \{0, 1, 2, \dots, 10\}$. Find:

1. $A^{\complement}$

2. $B^{\complement}$

3. $A \cap A^{\complement}$

4. $A \cup A^{\complement}$

5. $A - A^{\complement}$

6. $A - B^{\complement}$

[4] I believe the subpoints in the original text are slightly different to the subpoints of 5.1, but if you can do the above, you can certainly do those from the book too.

[5] If $0 \in \mathbb{N}$, then the set contains 0 as well.

7. $A^{\complement} - B^{\complement}$
8. $A^{\complement} \cap B$
9. $(A^{\complement} - B)^{\complement}$

[Go to Solution]

Exercise 6.2. Let $A = \{0, 2, 4, 6, 8\}, B = \{1, 3, 5, 7\}$ have universal set $U = \{0, 1, 2, \ldots, 8\}$. Find:

1. $A^{\complement}$
2. $B^{\complement}$
3. $A \cap A^{\complement}$
4. $A \cup A^{\complement}$
5. $A - A^{\complement}$
6. $(A \cup B)^{\complement}$
7. $A^{\complement} \cap B^{\complement}$
8. $(A \cap B)^{\complement}$
9. $A^{\complement} \times B$

[Go to Solution]

Exercise 6.3. Sketch the set $X = [1, 3] \times [1, 2]$ on the plane $\mathbb{R}^2$. On separate drawings, shade in the sets $X^{\complement}$ and $X^{\complement} \cap ([0, 2] \times [0, 3])$.

[Go to Solution]

Exercise 6.4. Sketch the set $X = [-1, 3] \times [0, 2]$ on the plane $\mathbb{R}^2$. On separate drawings, shade in the sets $X^{\complement}$ and $X^{\complement} \cap ([-2, 4] \times [-1, 3])$.

[Go to Solution]

Exercise 6.5. Sketch the set $X = \{(x, y) \in \mathbb{R}^2 : 1 \leq x^2 + y^2 \leq 4\}$ on the plane $\mathbb{R}^2$. On separate drawings, shade in the sets $X^{\complement}$

[Go to Solution]

Exercise 6.6. Sketch the set $X = \{(x, y) \in \mathbb{R}^2 : y < x^2\}$ on the plane $\mathbb{R}^2$. Shade in the sets $X^{\complement}$

[Go to Solution]

7 Venn Diagrams

My notes on this section are quite short since Venn Diagrams are only useful for the simpler set-theoretic operations. The important takeaways are:

1. If an expression uses only $\cup$, then parenthesis are optional. Similarly for $\cap$.
2. However, if an expression uses both $\cap, \cup$, then parenthesis are essential.

The exercises from this section will be skipped.

8 Indexed Sets

When a mathematical problem involves a lot of sets (which are almost all advanced problems), it is convenient to keep track of all of them by using *subscripts or indices.*

For example, let A, B, C be some sets, instead we can write A_1, A_2, A_3, and those would be the *indexed sets.*

Moreover, we can add the union and intersection operations and combine multiple sets. Consider the following rough guidelines.

Let $A_1, A_2, \ldots, A_n$ be sets, then $A_1 \cup A_2 \cup \cdots \cup A_n$ forms their union. We can say that the union consists of *at least one* of the sets A_i. The symmetric holds for intersections, i.e. $A_1 \cap A_2 \cap \cdots \cap A_n$ forms the intersection that consists of eveyrthing that is common among *all* the sets A_i. The following definitions follo:

Definition 8.0.1. Let $A_1, A_2, \ldots, A_n$ be sets, then:

$$A_1 \cup A_2 \cup \cdots \cup A_n = \{x : x \in A_i \text{ for at least one set } A_i, \text{ for } 1 \leq i \leq n\}$$
$$A_1 \cap A_2 \cap \cdots \cap A_n = \{x : x \in A_i \text{ for all sets } A_i, \text{ for } 1 \leq i \leq n\}.$$

If the number n is very large, then expression can get a bit messy. So we use the following notation similar to $\sum$-notation.

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$$
$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$

Example 8.0.1. Suppose $A_1 = \{0, 2, 5\}, A_2 = \{1, 2, 5\}, A_3 = \{2, 5, 7\}$. Then:

$$\bigcup_{i=1}^{3} A_i = A_1 \cup A_2 \cup A_3 = \{0, 1, 2, 5, 7\}$$
$$\bigcap_{i=1}^{3} A_i = A_1 \cap A_2 \cap A_3 = \{2, 5\}.$$

We can even use this notation when the list of sets A_i is infinite.

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup \cdots = \{x : \exists i \geq 1, x \in A_i\}$$
$$\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap \cdots = \{x : \forall i \geq 1, x \in A_i\}.$$

Example 8.0.2. Let $A_1 = \{-1, 0, 1\}, A_2 = \{-2, 0, 2\}, A_3 = \{-3, 0, 3\}, \ldots, A_i = \{-i, 0, i\}$. Observe that $\bigcup_{i=1}^{\infty} A_i = \mathbb{Z}, \bigcap_{i=1}^{\infty} A_i = \{0\}$.

A useful *heuristic* might be the following:

$$\bigcup_{i=1}^{3} = \bigcup_{i \in \{1,2,3\}} A_i$$

The idea here is that the union of the sets A_i, for $i = 1, 2, 3$ is the same as the intersection of A_i for i a member of the set $\{1, 2, 3\}$. Similarly,

$$\bigcup_{i=1}^{3} A_i = \bigcap_{i \in \{1,2,3\}} A_i$$
$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i \in \mathbb{N}} A_i$$
$$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i \in \mathbb{N}} A_i.$$

In general, the way this works is that we have a collection of sets A_i and $i \in I$ for I an index set. Furthermore, we can subscript anything we want really, so the following definition is more general than the previous one and might be helpful.

Definition 8.0.2. If A_α is a set for every α in some index set $I \neq 0$, then:

$$\bigcup_{\alpha \in I} A_\alpha = \{x : \exists \alpha \in I, x \in A_\alpha\}$$
$$\bigcap_{\alpha \in I} A_\alpha = \{x : \forall \alpha \in I, x \in A_\alpha\}.$$

8.1 Exercises for Section 8

Exercise 8.1. Suppose $A_1 = \{a, b, d, e, g, f\}, A_2 = \{a, b, c, d\}, A_3 = \{b, d, a\}, A_4 = \{a, b, h\}$. Find:

$$(1) \bigcup_{i=1}^{4} A_i = \dots \qquad (2) \bigcap_{i=1}^{4} A_i = \dots$$

[Go to Solution]

Exercise 8.2. Suppose $A_1 = \{0, 2, 4, 8, 10, \dots, 24\}, A_2 = \{0, 3, 6, 9, \dots, 24\}, A_3 = \{0, 4, 8, 12, \dots, 24\}$. Find:

$$(1) \bigcup_{i=1}^{3} A_i = \dots \qquad (2) \bigcap_{i=1}^{3} A_i = \dots$$

[Go to Solution]

Exercise 8.3. For each $n \in \mathbb{N}$, let $A_n = \{0, 1, 2, 3, \dots, n\}$. Find:

$$(1) \bigcup_{i \in \mathbb{N}} A_i = \dots \qquad (2) \bigcap_{i \in \mathbb{N}} A_i = \dots$$

[Go to Solution]

Exercise 8.4. For each $n \in \mathbb{N}$, let $A_n = \{-2n, 0, 2n\}$. Then find:

$$(1) \bigcup_{i \in \mathbb{N}} A_i \qquad (2) \bigcap_{i \in \mathbb{N}} A_i$$

[Go to Solution]

Exercise 8.5. Find:

$$(1)\ \bigcup_{i\in\mathbb{N}}[i,i+1] \qquad (2)\ \bigcap_{i\in\mathbb{N}}[i,i+1]$$

[Go to Solution]

Exercise 8.6.

$$(1)\ \bigcup_{i\in\mathbb{N}}[0,i+1] \qquad (2)\ \bigcap_{i\in\mathbb{N}}[0,i+1]$$

[Go to Solution]

Exercise 8.7.

$$(1)\ \bigcup_{i\in\mathbb{N}}\mathbb{R}\times[i,i+1] \qquad \bigcap_{i\in\mathbb{N}}\mathbb{R}\times[i,i+1]$$

[Go to Solution]

Exercise 8.8.

$$\bigcup_{a\in\mathbb{R}}\{a\}\times[0,1] \qquad \bigcap_{a\in\mathbb{R}}\{a\}\times[0,1]$$

[Go to Solution]

Exercise 8.9.

$$(1)\ \bigcup_{X\in\mathcal{P}(\mathbb{N})}X \qquad (2)\ \bigcap_{X\in\mathcal{P}(\mathbb{N})}X$$

[Go to Solution]

Exercise 8.10.

$$(1)\ \bigcup_{x\in[0,1]}[x,1]\times[0,x^2] \qquad (2)\ \bigcap_{x\in[0,1]}[x,1]\times[0,x^2]$$

[Go to Solution]

Exercise 8.11. Is $\bigcap_{\alpha\in I}A_\alpha\subseteq\bigcup_{\alpha\in I}A_\alpha$ always true for any collection of sets A_α with index set I?

[Go to Solution]

Exercise 8.12. If $\bigcap_{\alpha\in I}A_\alpha=\bigcup_{\alpha\in I}A_\alpha$, what do you think can be said about the relationship between the sets A_α?

[Go to Solution]

Exercise 8.13. If $J\neq\varnothing$ and $J\subseteq I$, does it follow that $\bigcup_{\alpha\in J}A_\alpha\subseteq\bigcup_{\alpha\in I}A_\alpha$? What about the respective intersections?

[Go to Solution]

Exercise 8.14. If $J\neq\varnothing, J\subseteq I$, does it follow that $\bigcap_{\alpha\in I}A_\alpha\subseteq\bigcap_{\alpha\in J}A_\alpha$? Explain.

[Go to Solution]

9 Sets That are Number Systems

In practice set theorists are most interested in the sets that have special properties and structures, such as $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$. We are assuming that all these sets are ordered. To illustrate it through an example, let $A \subseteq \mathbb{N}, A \neq \varnothing$, then because $\mathbb{N}$ is ordered, $x_0 \in A$ is going to be the smallest element in A. Similarly, $b \in \mathbb{Z}, A \subseteq \{b, b+1, b+2, \dots\}$. This principle is often called **The Well-Ordering Principle (WOP)**.
WOP fails for $\mathbb{R}$ and in general for $\mathbb{Z}, \mathbb{Q}$.
A consequence of WOP is that any integer a can be divided by a non-zero integer b, resulting in a quotient q and a remainder r. This is called the **division algorithm**.

Fact 9.0.1. (Division Algorithm) Given integers $a, b, b > 0$, there is some unique integer q, r for which $a = qb + r$ and $0 \leq r < b$.

The author of the textbook also explain how numbers can be understood in terms of sets.

$$0 = \varnothing, 1 = \{\varnothing\} = \{0\}, 2 = \{\varnothing, \{\varnothing\}\} = \{0, 1\}, \dots$$

In general, the natural number n is the set $n = \{0, 1, 2, 3, 4 \dots, n-1\}$ of the n numbers that come before it.

10 Russell's Paradox

This is a very important paradox in set theory and a lot has been written on it. The gist is the following.

$$A = \{X : X \text{ is a set and } X \notin X\}.$$

The idea here is to think of A as the *universe*. Then most sets will be members of it. Consider for example the set of integers $\mathbb{Z}$, we know that $Z \subseteq Z$ but $\mathbb{Z} \notin \mathbb{Z}$. The paradox arises from the question of whether A is an element of itself. So, for $X \in A, X \notin X$. If we suppose that $X = A$, we must have that $A \in A, A \notin A$. Therefore, $A \in A \Rightarrow A \notin A$ or vice versa. So, furthermore, we have $A \in A \Leftrightarrow A \notin A$. This leads to a contradiction. Russsell's Paradox instigated a very careful examination of set theory and evaluation of what can and cannot be regarded as a set. Eventually, the ZF axioms were developed and agreed upon.

Part II
Fundamentals Of Logic

This is a quick introduction to the chapter. The idea behind this chapter is to explore logic. Logic is the discipline or the process that teaches us how to deduce information correctly. It will be an invaluable tool when we reach Theorem proving. When we are proving theorems we apply logic to the information we are considering or to information that is already known to us, via some other theorems or lemmas. If our information is correct, then so is the information we deduced (the result).

11 Statements

Statements are *sentences* or mathematical expressions that are either true or false. There are statements in mathematics about which we might not yet know the truth and falsity of. These are called conjectures. However, once we discover whether they are true or false, they tend to remain that way. The idea is that no in between notion exists. The fact that we do not know something does not require its own truth-value.[6]

Example 11.0.1. These are some examples of *true* statements:

1. $2 \in \mathbb{Z}$
2. Some right angle triangles are isosceles.
3. Every even number is divisible by 2.

Example 11.0.2. Non-statements are paired with similar statements:

NOT a Statement	Statement
$\mathbb{Z}$	$42 \in \mathbb{Z}$
42	42 is not a number.
What is the solution of $2x = 84$?	The solution of $2x = 84$ is $x = 42$.

Example 11.0.3. We often use the letters $P, Q, R, S, \ldots$ to denote specific statements. When more letters are needed we tend to use subscripts.

1. $P = \forall n(n \in \mathbb{Z} \wedge n > 1) \Rightarrow 2^n - 1$ is prime.
2. Q : Every polynomial of degree n has at most n roots.
3. R : The function $f(x) = x^2$ is continuous.
4. $S_1 : \mathbb{Z} \subseteq \varnothing$.
5. $S_2 : \{0, -1, -2\} \cap \mathbb{N} = \varnothing$

Examples $(1) - (3)$ are true, and example $(4) - (5)$ are false.

Consider the following:

The integer x is even.

Call it $Q(x)$. Is this a statement? Not really. Since there is no definite truth value. We do not know what x is. Maybe it is a real number or a natural number. Only after we give a bit more information, would $Q(x)$ become a statement. As it stands, the above is an *open sentence* or an *open formula*.
Some statements are so important that they get their own name. For example

[6] Some people would heavily disagree with this. There is constructive mathematics and constructive logic, but for the time being we will just assume the above is true and roll with it. This 2-value truth-functionality is called *classical logic*.

Theorem 11.1.

$$R = \forall a, b, c, n \in \mathbb{N}(n > 2) \Rightarrow a^n + b^n \neq c^n.$$

Just because some statement is a theorem does not mean it is *obviously true* or *trivially true*. An example of one such statement is *Goldbach's Conjecture*, which states that every even integer greater than 2 is the sum of two primes. Currently, no such proof exists, unless OpenAI have published it, already...

11.1 Exercises For Section 11

Exercise 11.1. Decide whether or not the following are statements. If they are, decide whether if it is true or false, where possible.

1. Every real number is an even integer.
2. Every even integer is a real number.
3. If $x, y \in \mathbb{R}$ and $5x = 5y$, then $x = y$.
4. Sets $\mathbb{Z}$ and $\mathbb{N}$.
5. Sets $\mathbb{Z}$ and $\mathbb{N}$ are infinite.
6. Some sets are finite.
7. The derivative of any polynomial of degree 5 is a polynomial of degree 6.
8. $\mathbb{N} \notin \mathcal{P}(\mathbb{N})$.
9. $\cos(x) = -1$.
10. $(\mathbb{R} \times \mathbb{N}) \cap (\mathbb{N} \times \mathbb{R}) = \mathbb{N}^2$.
11. The integer x is a multiple of 7.
12. If the integer x is a multiple of 7, then it is divisible by 7.
13. Either x is a multiple of 7, or it is not.
14. Call me Ishamel.
15. In the beginning, God created the heaven and the earth.

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12 And, Or, Not

'And' conjoins two statements and produces a third one. When the two individual statements are both true, so is their conjuntion. Otherwise we say that the conjunction is false.

Suppose we have some propositional variables P, Q and their conjunction $P \wedge Q$,then:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Another way to conjoin sentences together would be through disjunction, in other words using 'or':

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

The requirement for a disjunction to be true is that *at least one* of the disjuncts is true. It is important to note here, that in mathemathics, we rarely use *inclusive* 'or'. This corresponds to 'either...or' but not both. If we ever want to express that we can, and it looks thus:

$$(P \vee Q) \wedge \neg Q,$$

for P, Q some statements. Then, notice how the truth table differs from that of *exclusive* disjunction.

P	Q	$\neg Q$	$(P \vee Q) \wedge \neg Q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

In the above example we can also see how the negation operator functions. At this point it is enough to say that it simply reverses the truth value of the propositional variable. However, later some important equivalences will become apparaent, when we look at De Morgan's Law or Conditional statements.

12.1 Exercises for Section 12

Exercise 12.1. Express each statement or open sentence in symbolic form. State what the propositional variables represent. You can use the following symbols $P, Q, \wedge, \vee, \neg, \in, =$

1. The number 8 is both even and a power of 2.
2. The matrix A is not invertible.
3. $x \neq y$
4. $x < y$
5. $y \geq x$
6. There is a quiz scheduled for Wednesday or Friday.
7. The number x equals zero, but the number y does not.
8. At least one of the numbers x and y equals 0.
9. $x \in A - B$.
10. $x \in A \cup B$.
11. $A \in \{X \in \mathcal{P}(\mathbb{N}) : |X^{\complement}| < \infty\}$.
12. Happy families are all alike, but each unhappy family is unhappy in its own way.
13. Human beings want to be good, but not too good, and not all the time.
14. A man should look for what is, and not for what he thinks should be.

[Go To Solution]

13 Conditional Statements

Consider the following statement:

R: If the integer a is a multiple of 6, then a is divisible by 2.

The statement can be broken down into two parts:

P: The integer is a multiple of 6 and Q: The integer is divisble by 2.

So, R becomes $P \Rightarrow Q$. This is a classic example of a conditional statement. Now, let us look at the truth table for the conditional operator.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

There are some peculiarities at first sight. Consider the example:

If you pass the exam, you will pass the course.

If were made the promise that if you pass the exam (second line of the table above), then you will pass the course (but you do not), something is wrong. Someone lied to you. The idea as to why the third case looks the way it does is somewhat complicated. I will try to explain it to the best of my ability. Consider the example:

If it is sunny outside, then people are happy.

It is easy to imagine that people are happy when it is raining or when it is cloudy. So the antecedent (the first part of the conditional statement) does not *strictly imply* the conclusion. To make sense of this, it is better to think about the antecedent as a type of hypothesis. A researcher can sometimes get the desired result with a wrong hypothesis. On a more formal level, there is a notion of *minimal logic* (which at this point might or might not be familiar to whoever is reading this) but the general point is that in order to construct Propositional Logic, we only require $\neg, \vee$ operators. Every other operator can be derived by those two. So, $P \Rightarrow Q$ becomes $\neg P \vee Q$. And this perfectly fits with the third case in the truth table, either the hypothesis is wrong or the conclusion is true but we cannot have that $P \vee \neg Q$, or that the hypothesis is true and the conclusion is logically false.[7]

Now, some alternative constructions of conditionals which is good to internalise:

1. If P, then Q.
2. Q if P.
3. Q whenever P.
4. Whenever P, then also Q.
5. P is a sufficient condition for Q.
6. For Q, it is sufficient that P.
7. Q is a necessary condition for P.
8. For P, it is necessary that Q.
9. P only if Q.

13.1 Exercises for Section 13

Exercise 13.1. Without changing their meaning, convert each of the following sentences into a sentences having the form 'if P, then Q'.

1. A matrix is invertible provided that its determinant is not zero.
2. For a function to be continuous, it is sufficient that it is differentiable.
3. For a function to be continuous, it is necessary that it is integrable.
4. A function is rational if it is a polynomial.

[7] I hope this sheds some light on the discussion. One can also simply choose the accept that this is true and later work out why.

5. An integer is divisible by 8 only if it is divisible by 4.
6. Whenever a surface has only one side, it is non-orientable.
7. A series converges whenever it converges absolutely.
8. A geometric series with ratio r converges if $|r| < 1$.
9. A function is integrable provided the function is continuous.
10. The discriminant is negative only if the quadratic equation has no real solutions.
11. You fail only if you stop writing.
12. People will generally accept facts as truth only if the facts agree with what they already beleive.
13. Whenever people agree with me I feel I must be wrong.

[Go to Solution]

14 Biconditional Statements

It is important to make a distinction between $P \Rightarrow Q$ and $Q \Rightarrow P$. We say that when $P \Rightarrow Q$, then $Q \Rightarrow P$ is its *converse*. However, when both statements are true we say that it is a *biconditional*.

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P) := P \Leftrightarrow Q$$

In terms of the *semantics* or, in other words, the truth table of the biconditional, the following is true:

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

We see that the biconditional is correct *only when* $P = T$ and $Q = T$ or when they both equal F.
There are many ways of saying 'if and only if' in English, some are as follows:

1. P iff Q.
2. P is a necessary and sufficient condition for Q.
3. P is equivalent to Q.
4. For P it is necessary and sufficient that Q.
5. If P, then Q and conversely.

14.1 Exercises for Section 14

Exercise 14.1. Without changing their meanings, convert each of the following sentences into a sentence having the form: $P \Leftrightarrow Q$.

1. For a matrix A to be invertible, it is necessary and sufficient that $\det(A) \neq 0$.
2. If a function has a constant derivative, then it is linear, and conversely.
3. If $xy = 0$, then $x = 0$ or $y = 0$, and conversely.
4. If $a \in \mathbb{Q}$, then $5a \in \mathbb{Q}$, and if $5a \in \mathbb{Q}$, then $a \in \mathbb{Q}$.
5. For an occurence to become an adventure, it is necessary and sufficient for one to recount it.

[Go to Solution]

15 Truth Tables for Statements

Why is it that a mathematical statement can have false truth values in its semantic representaion? Suppose the following:

> You will pass your exam iff you get an A or you get a B on your finals.

If you got an A and you did not pass the module, then you were lied to which is also reflected in the truth table as a 'false' truth value. The moral of the story is, people can lie but mathematics cannot! Trust no one!

P	Q	R	$P \vee (Q \Rightarrow R)$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	**F**
F	F	T	T
F	F	F	T

The truth table shows us that when $P = F$ and $R = F$, then the statement will be false.
I want to make a somewhat important note here for bigger truth tables. If you have never done big truth tables before, you might end up missing some cases. The best way I have found to make sure I have covered all the diferent cases (when doing a truth table manually) is the following.

- Always check how many propositional statements you have.
- If you have 3, for example, then you want to make sure you have 8 rows.
- 2^n = the number of rows your truth table needs, where n is the number of propositional statements.

- When writing out the truth table, I always start by writing the truth values for the first statement. Notice that, when there are 3 propositional values, the first one always has 4 Ts, followed by 4 Fs. The second has 2 Ts, followed by 2 Fs, followed by 2 Ts, and again by 2 Fs. And the last one has interchaging Ts and Fs on every row, starting with a T and finishing on an F. I have found this to work for me but there are many other ways to do it.
- This can be scaled to larger examples up to a point. For 4 propositional variables we would have 8 Ts, followed by 8 Fs, then 4 Ts, ... and so on.

15.1 Exercises for Section 15

Exercise 15.1. Write a truth table for the logical statements in problems $(1) - (9)$:

1. $P \vee (Q \Rightarrow R)$
2. $(Q \vee R) \Leftrightarrow (R \wedge Q)$
3. $\neg(P \Rightarrow Q)$
4. $\neg(P \vee Q) \vee (\neg P)$
5. $(P \wedge \neg P) \vee Q$
6. $(P \wedge \neg P) \wedge Q$
7. $(P \wedge \neg P) \Rightarrow Q$
8. $P \vee (Q \wedge \neg R)$
9. $\neg(\neg P \vee \neg Q)$

[Go to Solution]

Exercise 15.2. Suppose the statement $((P \wedge Q) \vee R) \Rightarrow (R \vee S)$ is false. Find the truth values of P, Q, R, S. (This can be done without a truth table, but try both)

[Go to Solution]

Exercise 15.3. Suppose P is false and the statement $(R \Rightarrow S) \Leftrightarrow (P \wedge Q)$ is true. Find the truth values of R, S. (Similarly, to the above exercise).

[Go to Solution]

16 Logical Equivalences

Logical equivalences refer to the notion of two formulas having the same truth values. Consider

$$P \Leftrightarrow Q \text{ and } (P \wedge Q) \vee (\neg P \wedge \neg Q)$$

or

$$P \Rightarrow Q \text{ and } \neg Q \Rightarrow \neg P.$$

Some equivalences are more important than others though. That is to say, some equivalences have their own special names. One such, is called *De Morgan's Law*:

1. $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
2. $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

The symbol '$\equiv$' simply states that these two expressions are equivalent. You might see me inconsistenly using $=$ and $\equiv$. For the time being they serve the same purpose.

Other such laws are the *commutative, associative, and distributive laws*:

Commutative $(P \wedge Q) \equiv (Q \wedge P)$.

Associative $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$.

Distributive $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$.

It is important to note here, that $\vee$ is also commutative and associative. If you have never seen these notions before, simply remember that same operators *associate with each other* whereas different operators *distribute over each other*. We can draw further comparisons from algebra to make sense of these. Another example of distribution is:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

and an example of association is:

$$x + (y + z) = x + y + z.$$

These are more familiar, but the idea is the exact same with propositional variables. One can draw further comparisons between $\vee$ and $\cup$ and $+$, as well as $\wedge, \cap, \cdot$. These share some structural similarities and this might help you understand them better, but they are not the same and they are not interchangebale. If you are doing a logic problem and you use $\cup$ instead of $\vee$, then this would be wrong.

16.1 Exercises for Section 16

Exercise 16.1. Use truth tables to show that the following statements are logically equivalent.

1. $P \wedge (Q \vee R) = (P \wedge Q) \vee (PlandR)$
2. $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$
3. $P \Rightarrow Q = \neg P \vee Q$
4. $\neg(P \vee Q) = \neg P \wedge \neg Q$
5. $\neg(P \vee Q \vee R) = \neg P \wedge \neg Q \wedge \neg R$
6. $\neg(P \wedge Q \wedge R) = \neg P \vee \neg Q \vee \neg R$
7. $P \Rightarrow Q = (P \wedge \neg Q) \Rightarrow (Q \wedge \neg Q)$
8. $\neg P \Leftrightarrow Q = (P \Rightarrow \neg Q) \wedge (\neg Q \Rightarrow P)$

[Go to Solution]

Exercise 16.2. Decide whether or not the following pairs of statements are logically equivalent. (I suggest to try and show these equivalences by reasoning through the problems and then check with truth tables if your answer was correct..)

1. $P \wedge Q$ and $\neg(\neg P \vee \neg Q)$
2. $(P \Rightarrow Q) \vee R$ and $\neg((P \wedge \neg Q) \wedge \neg R)$
3. $\neg P \wedge (P \Rightarrow Q)$ and $\neg(Q \Rightarrow P)$
4. $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$
5. $P \vee (Q \wedge R)$ and $(P \vee Q) \wedge R$
6. $P \wedge (Q \vee \neg Q)$ and $\neg P \Rightarrow (Q \wedge \neg Q)$

[Go to Solution]

Part III
Answers to All Exercises

Solutions to Exercises From Section 1

Solution. 1.1

1. $\{5x-1 : x \in \mathbb{Z}\} = \{\dots, -6, -1, 4, 9, \dots\}$.
2. $\{3x+2 : x \in \mathbb{Z}\} = \{\dots, -1, 2, 5, 8, \dots\}$.
3. $\{x \in \mathbb{Z} : -2 \leq x \leq 7\} = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$.
4. $\{x \in \mathbb{N} : -2 < x \leq 7\} = \{0, 1, 2, 3, 4, 5, 6, 7\}$.
5. $\{x \in \mathbb{R} : x^2 = 3\} = \{-\sqrt{3}, \sqrt{3}\}$.
6. $\{x \in \mathbb{R} : x^2 = 9\} = \{-3, 3\}$.
7. $\{x \in \mathbb{R} : x^2 + 5x = -6\} = \{-2, -3\}$.
8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\} = \{-3, -2, 0\}$.
9. $\{x \in \mathbb{R} : \sin \pi x = 0\} = \mathbb{Z}$.
10. $\{x \in \mathbb{R} : \cos x = 1\} = \{2n\pi : n \in \mathbb{Z}\}$.
11. $\{x \in \mathbb{Z} : |x| < 5\} = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$.
12. $\{x \in \mathbb{Z} : |2x| < 5\} = \{-2, -1, 0, 1, 2\}$.
13. $\{x \in \mathbb{Z} : |6x| < 5\} = \{0\}$.
14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\} = \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$.
15. $\{5a + 2b : a, b \in \mathbb{Z}\} = \mathbb{Z}$.
16. $\{6a + 2b : a, b \in \mathbb{Z}\} = 2\mathbb{Z}$.

□

Solution. 1.2 *Tip:* Some problems might look intimidating but in general, the easiest way to complete a problem like this, would be to consider the following three steps.

1 Is there common denominator?

2 Is there a constant that can be factored out?

3 What remains and how can this be generalised?

With those tips in mind, we proceed to the solutions.

17 $\{2, 4, 8, 16, 32, 64\} = \{2^x : x \in \mathbb{N}\}$

18 $\{0, 4, 16, 36, 64, 100, \dots\} = \{x^2 : x \in \mathbb{N}, 2 \mid x\}$.

19 $\{\dots, -6, -3, 0, 3, 6, 9, 12, 15, \dots\} = \{3x : x \in \mathbb{Z}\}$.

20 $\{\dots, -8, -3, 2, 7, 12, 17, \dots\} = \{5x + 2 : x \in \mathbb{Z}\}$.

21 $\{0, 1, 4, 9, 16, 25, 36, \dots\} = \{x^2 : x \in \mathbb{N}\}$

22 $\{3, 6, 11, 18, 27, 38, \dots\} = \{x^2 + 2 : x \in \mathbb{Z}^+\}$

23 $\{3, 4, 5, 6, 7, 8\} = \{x \in \mathbb{N} : 3 \leq x \leq 8\}$

24 $\{-4, -3, -2, -1, 0, 1, 2\} = \{x \in \mathbb{Z} : -4 \leq x \leq 2\}$

25 $\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\} = \{x \in \mathbb{Z} : 2^{-x}\}$

26 $\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27\} = \{3^{-x} : x \in \mathbb{Z}\}$

27 $\{\dots, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}\} = \{\frac{n\pi}{2} : n \in \mathbb{Z}\}$

28 $\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\} = \{\frac{3x}{4} : x \in \mathbb{Z}\}$

□

Solution. 1.3

29 $|\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}| = 3$

30 $|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}| = 5$

31 $|\{\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}\}| = 1$

32 $|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}\}| = 1$

33 $|\{x \in \mathbb{Z} : |x| < 10\}| = 19$

34 $|\{x \in \mathbb{N} : |x| < 10|\}| = 10$

35 $|\{x \in \mathbb{Z} : x^2 < 10\}| = 7$

36 $|\{x \in \mathbb{N} : x^2 < 10\}| = 4$

37 $|\{x \in \mathbb{N} : x^2 < 0\}| = 0$

38 $|\{x \in \mathbb{N} : 5x \leq 20\}| = 5.$

□

Solution. 1.4

39 $\{(x, y) : x \in [1, 2], y \in [1, 2]\} \Rightarrow$ This would be a solid 1×1 square in $\mathbb{R}^2$ with corners $(1, 1), (1, 2), (2, 1), (2, 2)$

40 $\{(x, y) : x \in [0, 1], y \in [1, 2]\} \Rightarrow$ This would be a solid 1×1 square in $\mathbb{R}^2$ with corners $(0, 1), (0, 2), (1, 1), (1, 2)$.

41 $\{(x, y) : x \in [-1, 1], y = 1\} \Rightarrow$ This would be a line connecting $(-1, 1)$ with $(1, 1)$.

42 $\{(x, y) : x = 2, y \in [0, 1]\} \Rightarrow$ This would be a line connecting $(2, 0), (2, 1)$

43 $\{(x, y) : |x| = 2, y \in [0, 1]\} \Rightarrow$ This would be two vertical line segments, that connect $(-2, 0) - (-2, 1)$ and the other one would be $(2, 0) - (2, 1)$.

44 $\{(x, x^2) : x \in \mathbb{R}\} \Rightarrow$ This would be the parabolla $y = x^2$.

45 $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\} \Rightarrow$ This would be the unit circle centered at the origin $(0, 0)$ with $r = 1$

46 $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 1\} \Rightarrow$ This would be a disk with $r = 1$, centered at $(0, 0)$.

47 $\{(x, y) : x, y \in \mathbb{R}, y \geq x^2 - 1\} \Rightarrow$ This would be the region underneath the parabola $y = x^2 - 1$.

48 $\{(x, y) : x, y \in \mathbb{R} x > 1\} \Rightarrow$ This would the whole upper right and down left quadrants of the xy-plane, or an *open half-plane*, starting from $x = 2$.

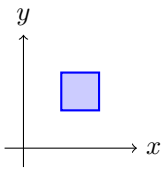
49 $\{(x, x + y) : x \in \mathbb{R}, y \in \mathbb{Z}\} \Rightarrow$ This would be infinitely many parallel lines with slope 1.

50 $\{(x, \frac{x^2}{y}) x \in \mathbb{R}, y \in \mathbb{N}\} \Rightarrow$ Division by 0 is undefined, so we skip those values. Notice that the relation between the first and second coordinate is quadratic, instead of linear. That is to say $v = \frac{u^2}{k}$, so we have a upward-opening parabolas passing through the origin

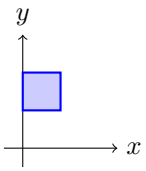
51 $\{(x, y) \in \mathbb{R}^2 : (y - x)(y + x) = 0\} \Rightarrow$ Here we have lines passing through the origin in an X shape.

52 $\{(x, y) \in \mathbb{R}^2 : (y - x^2)(y + x^2) = 0\} \Rightarrow$ Lastly, we have two parabolas touching at the origin one opening upwards, one downwards, $y = x^2, y = -x^2$, respectively.

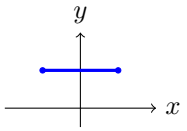
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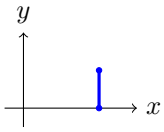
(a) Item 39



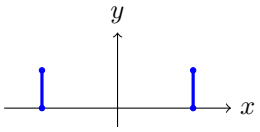
(b) Item 40



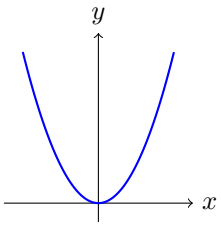
(c) Item 41



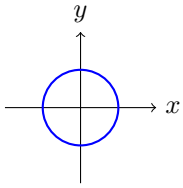
(d) Item 42



(e) Item 43

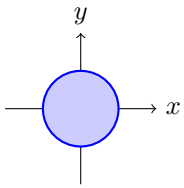


(f) Item 44

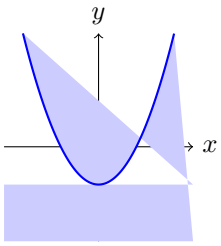


(g) Item 45

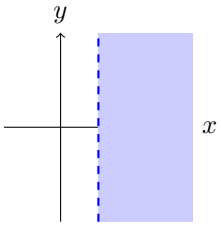
Figure 1: Geometric Representations 39–45



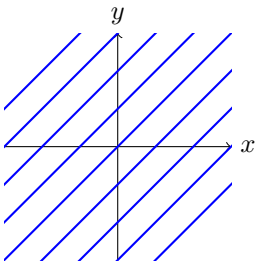
(h) Item 46



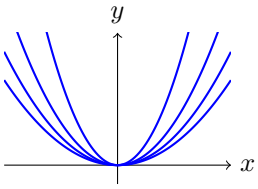
(i) Item 47



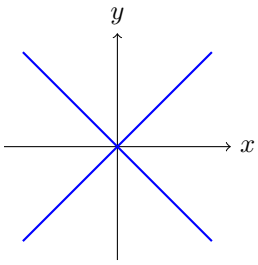
(j) Item 48



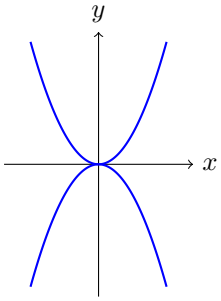
(k) Item 49



(l) Item 50



(m) Item 51



(n) Item 52

Figure 1: Geometric Representations 46–52 (Continued)

Solutions to Exercises from Section 2

Solution. 2.1

$(a) A \times B = \{(1,a),(1,c),(2,a),(2,c),(3,a),(3,c),(4,a),(4,c)\}$

$(b) B \times A = \{(a,1),(c,1),(a,2),(c,2),(a,3),(c,3),(a,4),(c,4)\}$

$(c) A \times A = A^2 = \{(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),$

$(2,4),(3,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,3),(4,4)\}$
$(d) B \times B = B^2 = \{(a,a),(a,c),(c,a),(c,c)\}$

$(e) \varnothing \times B = \{(a,b) : a \in \varnothing, b \in B\} = \varnothing$

$(f)(A \times B) \times B = \{(1,a),(1,c),(2,a),(2,c),(3,a),(3,c),(4,a),(4,c)\} \times \{a,c\} =$
$\{((1,a),a),((1,c),a),((2,a),a),$
$((2,c),a),((3,a),a),((3,c),a),$
$((4,a),a),((4,c),a),((1,a),c),$
$((1,c),c),((2,a),c),((2,c),c),$
$((3,a),c),((3,c),c),((4,a),c),((4,c),c)\}$

$(g) A \times (B \times B) = \{1,2,3,4\} \times \{(a,a),(a,c),(c,a),(c,c)\} =$
$\{(1,(a,a)),(1,(a,c)),(1,(c,a)),(1,(c,c)),$
$(2,(a,a)),(2,(a,c)),(2,(c,a)),(2,(c,c))\},$
$(3,(a,a)),(3,(a,c)),(3,(c,a)),(3,(c,c)),$
$(4,(a,a)),(4,(a,c)),(4,(c,a)),(4,(c,c))$

$(h) B^3 = \{(a,a,a),(a,a,c),(a,c,a),(a,c,c),(c,a,a),(c,a,c),(c,c,a),(c,c,c)\} = |2^3| = 8$

This ends Part I of 2.1. Now, for part II.

$(a) A \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\}$

$(b) B \times A = \{(0, \pi), (1, \pi), (0, e), (1, e), (0, 0), (1, 0)\}$

$(c) A^2 = \{(\pi, \pi), (\pi, e), (\pi, 0)$
$(e, \pi), (e, e), (e, 0)$
$(0, \pi), (0, e), (0, 0)\}$

$(d) B^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

$(e) A \times \varnothing = \varnothing$

$(f)(A \times B) \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\} \times \{0, 1\}$
$\{((\pi, 0), 0), ((\pi, 0), 1), ((\pi, 1), 0),$
$((\pi, 1), 1,), ((e, 0), 0), ((e, 0), 1),$
$((e, 1), 0), ((e, 1), 1), ((0, 0), 0),$
$((0, 0), 1), ((0, 1), 0), ((0, 1), 1)\}$

$(g) A \times (B \times B) = \{\pi, e, 0\} \times \{(0, 0), (0, 1), (1, 0), (1, 1)\}$
$\{(\pi, (0, 0)), (\pi, (0, 1)), (\pi, (1, 0)), (\pi, (1, 1)),$
$(e, (0, 0)), (e, (0, 1)), (e, (1, 0)), (e, (1, 1))$
$(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), (0, (1, 1))\}$

$(h) A \times B \times B = \{(\pi, 0, 0), (\pi, 0, 1), (\pi, 1, 0),$
$(\pi, 1, 1), (e, 0, 0), (e, 0, 1),$
$(e, 1, 0), (e, 1, 1), (0, 0, 0),$
$(0, 0, 1), (0, 1, 0), (0, 1, 1)\}$

This concludes part II. Lastly, part III.

(3) $\{x \in \mathbb{R} : x^2 = 2\} \times \{a, c, e\} =$
$\{-\sqrt{2}, \sqrt{2}\} \times \{a, c, e\} =$
$\{(-\sqrt{2}, a), (-\sqrt{2}, c), (-\sqrt{2}, e),$
$(\sqrt{2}, a), (\sqrt{2}, c), (\sqrt{2}, e)\}$

(4) $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\} =$
$\{3, 4\} \times \{-5, 5\} = \{(3, -5), (3, 5), (4, -5), (4, 5)\}$

(5) $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\} =$
$\{-\sqrt{2}, \sqrt{2}\} \times \{-2, 2\} = \{(-\sqrt{2}, -2), (-\sqrt{2}, 2), (\sqrt{2}, -2), (\sqrt{2}, 2)\}$

(6) $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\} =$
$\{-1, 0, 1\} \times \{0, 1\} =$
$\{(-1, 0), (-1, 1), (0, 0), (0, 1), (1, 0), (1, 1)\}$

(7) $\{\varnothing\} \times \{0, \varnothing\} \times \{0, 1\} =$
$\{(\varnothing, 0, 0), (\varnothing, \varnothing, 0), (\varnothing, 0, 1), (\varnothing, \varnothing, 1)\}$

(8) $\{0, 1\}^4 = \{(0, 0, 0, 0), (0, 0, 0, 1), (0, 0, 1, 0), (0, 1, 0, 0),$
$(1, 0, 0, 0), (0, 0, 1, 1), (0, 1, 0, 1), (1, 0, 0, 1),$
$(0, 1, 1, 0), (1, 1, 0, 0), (1, 1, 1, 0), (1, 1, 1, 1),$
$(0, 1, 1, 1), (1, 0, 1, 0), (1, 0, 1, 1), (1, 1, 0, 1)\}.$

□

Solution. 2.2 As in the previous exercise where we sketched out the sets on the xy-plane, we will

first solve them/explain them and then go on to sketch them.

(9)$\{1,2,3\} \times \{-1,0,1\} =$
$\{(1,-1),(1,0),(1,1),$
$(2,-1),(2,0),(2,1),$
$(3,-1),(3,0),(3,1)\}$

(10)$\{-1,0,1\} \times \{1,2,3\} =$
$\{(-1,1),(-1,2),(-1,3),$
$(0,1),(0,2),(0,3),$
$(1,1),(1,2),(1,3)\}$

(11)$[0,1] \times [0,1]$ these would be all the paired points in that interval starting from
$x = 0, y = 1$

(12)$[-1,1] \times [1,2]$ these would be all the paired points in that interval starting from
$x = -1, y = 1$ and filling the space up to $x = 1, y = 2$

(13)$\{1, 1.5, 2\} \times [1,2]$ similarly to the cases above, those will become more obvious when we sketch them out

(14)$[1,2] \times \{1, 1.5, 2\},$ ditto.

(15)$\{1\} \times [0,1]$

(16)$[0,1] \times \{1\}$

(17)$\mathbb{N} \times \mathbb{Z},$ here, we take the natural numbers as our first coordinate and the integers as our second.

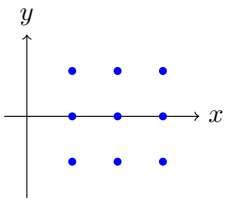
(18)$\mathbb{Z}^2,$ here we have the whole xy-plane with whole positive and negative integers.

(19)$[0,1]^2$

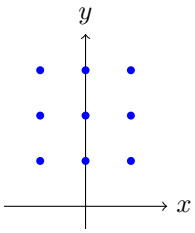
(20)$\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0,1].$

Now, we will show them plotted on the xy-plane.

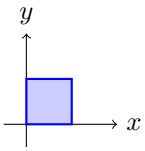
□



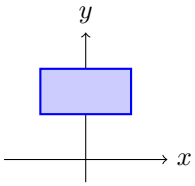
(a) Item (9)



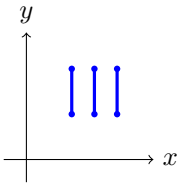
(b) Item (10)



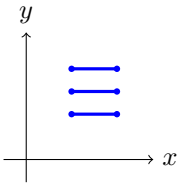
(c) Item (11)



(d) Item (12)

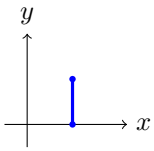


(e) Item (13)

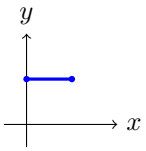


(f) Item (14)

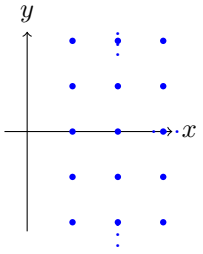
Figure 2: Plots for Cartesian Products (Items 9–14)



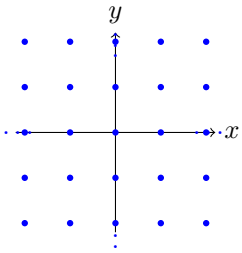
(g) Item (15)



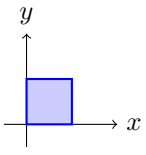
(h) Item (16)



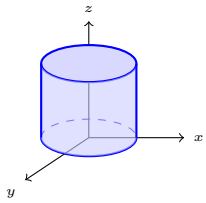
(i) Item (17) $\mathbb{N} \times \mathbb{Z}$



(j) Item (18) $\mathbb{Z}^2$



(k) Item (19) $[0,1]^2$



(l) Item (20) Disk $\times [0,1]$

Figure 2: Plots for Cartesian Products (Items 15–20, Continued)

Solutions to Exercises from Section 3

Solution. 3.1

(1) $\{1,2,3,4\} =$
$\varnothing, \{1\}, \{2\}, \{3\}, \{4\},$
$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\},$
$\{2,4\}, \{3,4\}, \{1,2,3\}, \{1,2,4\},$
$\{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}$

(2) $\{1,2,\varnothing\} =$
$\varnothing, \{1\}, \{2\},$
$\{\varnothing\}, \{1,2\}, \{1,\varnothing\},$
$\{2,\varnothing\}, \{1,2,\varnothing\}$

(3) $\{\{\mathbb{R}\}\} = \varnothing, \{\mathbb{R}\}$

(4) $\varnothing = \varnothing$

(5) $\{\varnothing\} = \varnothing, \{\varnothing\}$

(6) $\{\mathbb{R}, \mathbb{Q}, \mathbb{N}\} =$
$\varnothing, \{\mathbb{R}\}, \{N\}, \{\mathbb{Q}\},$
$\{\mathbb{R}, \mathbb{N}\}, \{\mathbb{R}, \mathbb{Q}\}, \{\mathbb{N}, \mathbb{Q}\}, \{\mathbb{R}, \mathbb{N}, \mathbb{Q}\}$

(7) $\{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\} =$
$\varnothing, \{\mathbb{R}\}, \{\{\mathbb{N}, \mathbb{Q}\}\}, \{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\}$

(8) $\{\{0,1\}, \{0,1,\{2\}\}, \{0\}\} =$
$\varnothing, \{\{0,1\}\}, \{\{0,1,\{2\}\}\}, \{\{0\}\},$
$\{\{0,1\}, \{0,1,\{2\}\}\}, \{\{0,1\}, \{0\}\},$
$\{\{0,1,\{2\}\}, \{0\}\}, \{\{0,1\}, \{0,1,\{2\}\}, \{0\}\}.$

□

Solution. 3.2

(1) $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 2\} = \{\{3, 2\}, \{3, a\}, \{2, a\}\}$

(2) $\{X \subseteq \mathbb{N} : |X| \leq 1\} = \{\varnothing\} \cup \{\{n\} : n \in \mathbb{N}\}$

(3) $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 4\} = \varnothing$

(4) $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 1\} = \{\{3\}, \{2\}, \{a\}\}$

□

Solution. 3.3

(1) $\mathbb{R}^3 \subseteq \mathbb{R}^3$

(1) is true, for the fact that every set is a subset of itself.

(2) $\mathbb{R}^2 \subseteq \mathbb{R}^3$

(2) is tricky, strictly speaking it is false. However, if you have any experience with Linear Algebra you might consider it true. In Linear Algebra we abuse the notation and say that $\mathbb{R}^3$ is an extension of $\mathbb{R}^2$, addiing the third dimension to a two dimensional plane. In set theory however, we are a bit more strict. The above is false, since

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R} \wedge y \in \mathbb{R}\},$$

And

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) : x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}\}.$$

So, the first *object* takes in pairs, and the second object takes in triples.

(3) $\{(x, y) \in \mathbb{R}^2 : x - 1 = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\}$.

Consider a small example here, the set contains pairs (x, y), the first element of the pair has to be 1, so $\{(1, y) : y \in \mathbb{R}\}$. In the second set we have $x^2 - x = 0$, solving this gives us $x = 0, \vee x = 1$, so we have the pairs $\{(0, y), (1, y) : y \in \mathbb{R}\}$. Then it follows that the example is true.

(4) $\{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x - 1 = 0\}$.

This example is false, in the LHS, we can have $0^2 - 0 = 0$, whereas in the RHS we cannot, unless $x = 1$. □

Solutions to Exercises from Section 4

Solution. 4.1 I will also show the cardinality of the powerset since I think it helps with counting all its subsets.

(1) $\mathcal{P}(\{\{a,b\},\{c\}\}) = \{\varnothing, \{\{a,b\}\}, \{\{c\}\}, \{\{a,b\},\{c\}\}\}$ $\quad |\mathcal{P}(\{\{a,b\},\{c\}\})| = 2^2 = 4$ elements

(2) $\mathcal{P}(\{1,2,3,4\}) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{4\},$
$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\},$
$\{3,4\}, \{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}\} = 2^4 = 16$ elements.

(3) $\mathcal{P}(\{\{\varnothing\},5\}) = \{\varnothing, \{\{\varnothing\}\}, \{5\}, \{\{\varnothing\},5\}\} = 2^2 = 4$ elements.

(4) $\mathcal{P}(\{\mathbb{R},\mathbb{Q}\}) = \{\varnothing, \{\mathbb{R}\}, \{\mathbb{Q}\}, \{\mathbb{R},\mathbb{Q}\}\} = 2^2 = 4$ elements.

(5) $\mathcal{P}(\mathcal{P}(\{2\})) = \mathcal{P}(\{\varnothing, \{2\}\}) =$
$\{\varnothing, \{\varnothing\}, \{\{2\}\}, \{\varnothing, \{2\}\}\} = 2^2 = 4$ elements.

(6) $\mathcal{P}(\{1,2\}) \times \mathcal{P}(\{3\}) =$
$\{\varnothing, \{1\}, \{2\}, \{1,2\}\} \times \{\varnothing, \{3\}\} =$
$\{(\varnothing,\varnothing), (\varnothing, \{3\}), (\{1\},\varnothing), (\{1\},\{3\}),$
$(\{2\},\varnothing), (\{2\},\{3\}), (\{1,2\},\varnothing), (\{1,2\},\{3\})\}$

(7) $\mathcal{P}(\{a,b\}) \times \mathcal{P}(\{0,1\}) =$
$\{\varnothing, \{a\}, \{b\}, \{a,b\}\} \times \{\varnothing, \{0\}, \{1\}, \{0,1\}\} =$
$\{(\varnothing,\varnothing), (\varnothing,\{0\}), (\varnothing,\{1\}), (\varnothing,\{0,1\}),$
$(\{a\},\varnothing), (\{a\},\{0\}), (\{a\},\{1\}), (\{a\},\{0,1\}),$
$(\{b\},\varnothing), (\{b\},\{0\}), (\{b\},\{1\}), (\{b\},\{0,1\}),$
$(\{a,b\},\varnothing), (\{a,b\},\{0\}), (\{a,b\},\{1\}), (\{a,b\},\{0,1\})\}$

(8) $\mathcal{P}(\{1,2\} \times \{3\}) = \mathcal{P}(\{(1,3),(2,3)\}) =$
$\{\varnothing, \{(1,3)\}, \{(2,3)\}, \{(1,3),(2,3)\}\}$

(9) $\mathcal{P}(\{a,b\} \times \{0\}) = \mathcal{P}(\{(a,0),(b,0)\}) =$
$\{\varnothing, \{(a,0)\}, \{(b,0)\}, \{(a,0),(b,0)\}\}$

We will do the last three examples separately, since there is a bit more to consider. Let's start with (10):

$$(10)\ \{X \in \mathcal{P}(\{1,2,3\}) : |X| \leq 1\}$$

Notice that we are looking for a *member* X, that has size no bigger than 1. So, let's count the powerset and then see which sets qualify.

$$\mathcal{P}(\{1,2,3\}) = \{\varnothing, \{1\}, \{2\}, \{3\},$$
$$\{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\} = 2^3 = 8\text{elements}$$

Now, for the condition $X \in \mathcal{P}(\{1,2,3\})$. We have

$$\{\varnothing, \{1\}, \{2\}, \{3\}\}.$$

These are all the singletons and the empty set. For (11), we have

$$\{\varnothing, \{\varnothing\}, \{\{1\}\}, \{\{2\}\}, \{\{3\}\}, \{\{1,2\}\}, \{\{1,3\}\}, \{\{2,3\}\}, \{\{1,2,3\}\}\}.$$

This might seem strange, but these are all of cardinality at most than 1. Notice that the subsets of the powerset are all enclosed in an extra pair of curly braces, so they are counted as having one element. Lastly, for (12) the condition changes slightly, we need to find $X \in \mathcal{P}(\{1,2,3\})$ such that $2 \in X$. So,

$$\{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}.$$

□

Solution. 4.2 We let $|A| = m, |B| = n$, then

(1) $|\mathcal{PPP}(A)| = 2^{2^{2^m}}$

(2) $|\mathcal{PP}(A)| = 2^{2^m}$

(3) $|\mathcal{P}(A \times B)| = 2^{m \cdot n}$

(4) $|\mathcal{P}(A) \times \mathcal{P}(B)| = 2^m \cdot 2^n$

(5) $|\{X \in \mathcal{P}(A) : |X| \leq 1\}| = m + 1$

(6) $|\mathcal{P}(A \times \mathcal{P}(B))| = 2^{m \cdot 2^n}$

(7) $|\mathcal{PPP}(A \times \varnothing)| = 4$

(8) $|\{X \subseteq \mathcal{P}(A) : |X| \leq 1\}| = 2^m + 1.$

Some Further Comments: For (5) it is important to understand the following. We are looking at members of the powerset, as in the previous exercise, we saw that these would be all the singletons and the empty set. Since we have exactly m singletons and we add the empty set to that, so we get $m + 1$. Since the product of any set with the empty set is empty, we have to calculate $|\mathcal{PPP}(\varnothing)|$. The idea in (8) is the same as in the previous exercise. □

Solutions to Exercises from Section 5

Solution. 5.1

1. $A \cup B = \{1, 3, 4, 5, 6, 7, 8, 9\}$
2. $A \cap B = \{4, 6\}$
3. $A - B = \{1, 3, 7, 9\}$
4. $A - C = \{1, 3, 6, 7, 9\}$
5. $B - A = \{5, 8\}$
6. $A \cap C = \{4\}$
7. $B \cap C = \{4, 5, 8\}$
8. $B \cup C = \{4, 5, 6, 8\}$
9. $C - B = \varnothing$

□

Solution. 5.2

1. $A \cup B = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$
2. $A \cap B = \varnothing$
3. $A - B = A$ since $A \cap B = \varnothing$ (or in other words are disjoint)
4. $A - C = \{0, 6\}$
5. $B - A = B$
6. $A \cap C = \{2, 8, 4\}$
7. $B \cap C = \varnothing$
8. $B \cup C = \{1, 2, 3, 4, 5, 7, 8\}$
9. $C - B = C$

□

Solutions to Exercises from Section 6

Solution. 6.1

1. $A^{\complement} = \{0, 2, 5, 8, 10\}$
2. $B^{\complement} = \{0, 1, 2, 3, 7, 9, 10\}$
3. $A \cap A^{\complement} = \{0, 2, 5, 8, 10\} \cap \{4, 3, 6, 7, 1, 9\} = \varnothing$
4. $A \cup A^{\complement} = U$ by the same logic as the above example
5. $A - A^{\complement} = \{0, 2, 5, 8, 10\} - \{4, 3, 6, 7, 1, 9\} = \{0, 2, 5, 8, 10\} = A$
6. $A - B^{\complement} = \{4, 6\}$
7. $A^{\complement} - B^{\complement} = \{5, 8\}$
8. $A^{\complement} \cap B = \{5, 8\}$
9. $(A^{\complement} - B)^{\complement} = \{1, 3, 4, 5, 6, 7, 9\}$

□

Solution. 6.2

1. $A^{\complement} = \{1, 3, 5, 7\} = B$
2. $B^{\complement} = \{0, 2, 4, 6, 8\} = A$
3. $A \cap A^{\complement} = \varnothing$
4. $A \cup A^{\complement} = U$
5. $A - A^{\complement} = A$
6. $(A \cup B)^{\complement} = \varnothing$
7. $A^{\complement} \cap B^{\complement} = \varnothing$
8. $(A \cap B)^{\complement} = U$
9. $A^{\complement} \times B = \{(1, 1), (1, 3), (1, 5), \dots\} = B^2$

□

Solution. 6.3 You can find the solution for this exercise below. □

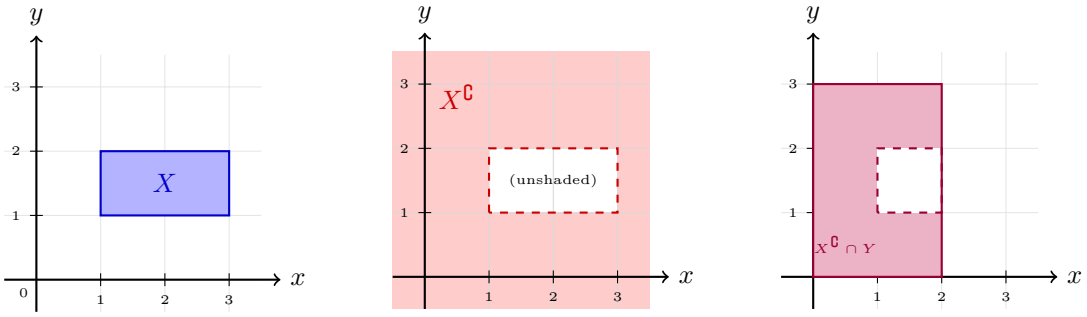
Solution. 6.4 Ditto.

□

Solution. 6.5 Ditto. □

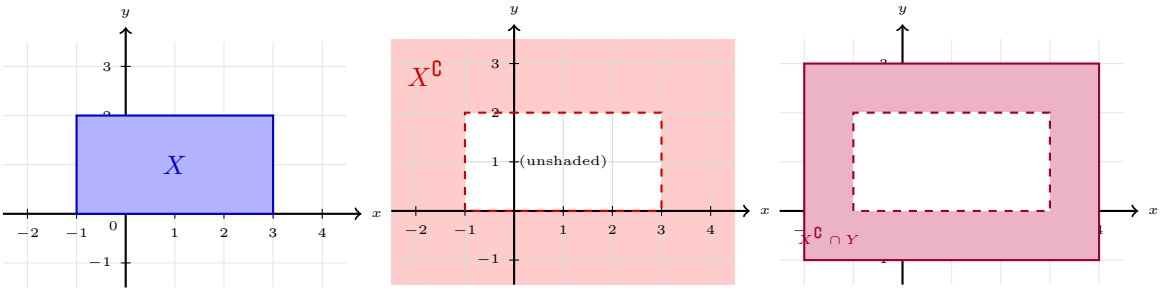
Solution. 6.6 Ditto.

□



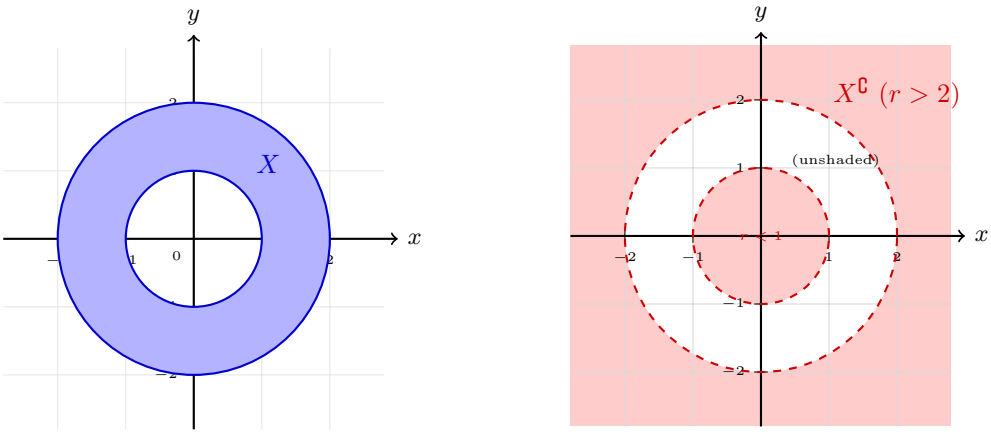
(a) The set $X = [1, 3] \times [1, 2]$ (b) The complement $X^{\complement}$ (c) $X^{\complement} \cap ([0, 2] \times [0, 3])$

Figure 3: Sketches of X, its complement $X^{\complement}$, and the restricted intersection.



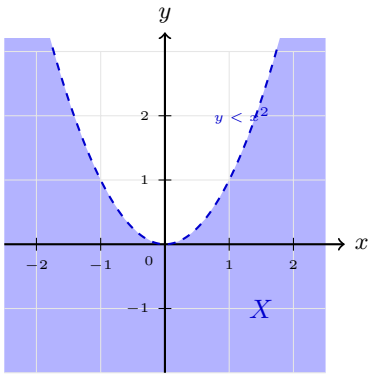
(a) The set X (b) The complement $X^{\complement}$ (c) $X^{\complement} \cap ([-2, 4] \times [-1, 3])$

Figure 4: Sketches for Exercise 6.4.

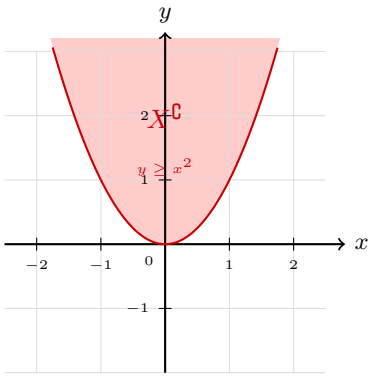


(a) The set X (Annulus $1 \leq r \leq 2$) (b) The complement $X^{\complement}$ $(r < 1 \cup r > 2)$

Figure 5: Sketches for Exercise 6.5.



(a) The set $X = \{(x, y) \in \mathbb{R}^2 : y < x^2\}$



(b) The complement $X^C = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$

Figure 6: Sketches for Exercise 6.6.

Solutions to Exercises from Section 7

Skipped.

Solutions to Exercises from Section 8

Solution. 8.1 Let $A_1 = \{a, b, d, e, g, f\}, A_2 = \{a, b, c, d\}, A_3 = \{b, d, a\}, A_4 = \{a, b, h\}$. Then,

$$(1) \bigcup_{i=1}^{4} A_i = A_1 \cup A_2 \cup A_3 \cup A_4 = \{a, b, c, d, e, f, g, h\}$$

and

$$(2) \bigcap_{i=1}^{4} A_i = A_1 \cap A_2 \cap A_3 \cap A_4 = \{a, b\}.$$

□

Solution. 8.2 Let $A_1 = \{0, 2, 4, \ldots, 24\}, A_2 = \{0, 3, 6, \ldots, 24\}, A_3 = \{0, 4, 8, \ldots, 24\}$. Then,

$$(1) \bigcup_{i=1}^{3} A_i = \{0, 2, 3, 4, 6, 9, \ldots, 24\}$$

and

$$(2) \bigcap_{i=1}^{3} A_i = \{0, 12, 14\}.$$

□

Solution. 8.3 For each $n \in \mathbb{N}$, we let $A_n = \{0, 1, 2, 3, 4, \ldots, n\}$. Let us first unpack what this means. Consider the following two interpretations:

$$A_0 = \{0\}, A_1 = \{0, 1\}, \ldots \qquad A_1 = \{0, 1\}, A_2 = \{0, 1, 2\}, \ldots$$

It is up to convention whether $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ or $\mathbb{N} = \{1, 2, 3, \ldots\}$. Depending on the convention the answer might differ slightly. Case 1:

$$(1) \bigcup_{i \in \mathbb{N}} A_i = \mathbb{N}$$

Case 2:

$$(1) \bigcup_{i \in \mathbb{N}} A_i = \mathbb{N} \cup \{0\}.$$

Similarly for the second subpoint. Case 1:

$$(2) \bigcap_{i \in \mathbb{N}} A_i = \{0, 1\}$$

and Case 2:

$$(2) \bigcap_{i \in \mathbb{N}} A_i = \{0\}$$

□

Solution. 8.4 Let each $n \in \mathbb{N}$ with $A_n = \{-2n, 0, 2n\}$, then similarly to before, we have

$$A_0 = \{-2(0), 0, 2(0)\} \qquad A_1 = \{-2(1), 0, 2(1)\}, A_2 = \{-2(2), 0, 2(2)\}$$

So,

$$\bigcup_{i \in \mathbb{N}} A_i = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$

and

$$\bigcap_{i \in \mathbb{N}} A_i = \{0\}$$

Similarly to before, if we assume that $\mathbb{N} = \mathbb{N} - \{0\}$, then we will start from A_1 but this exercise will not be affected by that convetion like the previous one. □

Solution. 8.5

$$(1) \bigcup_{i \in \mathbb{N}} [i, i + 1] = [0, \infty)$$

Here we assume that $0 \in \mathbb{N}$, if you choose to assume the contrary, then the interval would start with 1 instead of 0.

$$(2) \bigcap_{i \in \mathbb{N}} [i, i + 1] = \varnothing$$

Since some pairs of intervals are disjoint ($[0, 1] \cap [2, 3]$). □

Solution. 8.6

$$(1)\ \bigcup_{i\in\mathbb{N}}[0,i+1]=[0,\infty)$$

and

$$(2)\ \bigcap_{i\in\mathbb{N}}[0,i+1]=[0,1] \text{ or } [0,2] \text{ depending on whether we include } 0\in\mathbb{N}$$

□

Solution. 8.7

$$(1)\ \bigcup_{i\in\mathbb{N}}\mathbb{R}\times[i,i+1]=\mathbb{R}\times[0,\infty)$$

Again, in the above, convention matters, it will be the differnece between $[1,\infty)$ and the answer provided above.

$$(2)\ \bigcap_{i\in\mathbb{N}}\mathbb{R}\times[i,i+1]=\varnothing$$

□

Solution. 8.8

$$(1)\ \bigcup_{a\in\mathbb{R}}\{a\}\times[0,1]=\mathbb{R}\times[0,1]$$

We know that $a\in\mathbb{R}$ means that a is any real number such that $\sqrt{2}\times[0,1]$ for example.

$$(2)\ \bigcap_{a\in\mathbb{R}}\{a\}\times[0,1]=\varnothing$$

□

Solution. 8.9

$$(1)\ \bigcup_{X\in\mathcal{P}(\mathbb{N})}X=\mathbb{N}$$

The way to think about it is that because $X\in\mathcal{P}(\mathbb{N})$, then X can be any subset of the natural numbers. The union of those subsets would be the natural numbers.

$$(2)\ \bigcap_{X\in\mathcal{P}(\mathbb{N})}X=\varnothing$$

Important here is to remember that the empty set is part of $\mathcal{P}(\mathbb{N})$. □

Solution. 8.10

$$(1)\ \bigcup_{x\in[0,1]}[x,1]\times[0,x^2]=\{(u,v):0\leq u\leq 1,0\leq v\leq u^2\}$$

and

$$(2) \quad \bigcap_{x\in[0,1]} [x,1]\times[0,x^2] = \{(1,0)\}$$

□

Solution. 8.11 Yes, this is always true. Informally speaking, the intersection of two sets is almost always smaller than their union. Consider the example:

$$\{1,2,3\}\cup\{2,3,4\} = \{1,2,3,4\} \qquad \{1,2,3\}\cap\{2,3,4\} = \{2,3\}.$$

In the case when the intersection is empty and the two sets disjoint, then we know that the empty set is a subset of any set, so the statement still holds.

For a more formal proof, let $\alpha \in I$, then suppose for some $x \in \alpha_0$. Since, $x \in \alpha_0$ and $\alpha \in I$, then $x \in A_{\alpha_0}$. Therefore, $x \in \bigcup_{\alpha\in I} A_\alpha$. □

Solution. 8.12 Now we consider the relationship between $\bigcap_{\alpha\in I} A_\alpha = \bigcup_{\alpha\in I} A_\alpha$. Clearly if they are equal the first thing to consider would be that two sets are equal, meaning

$$\{1,2,3\}\cap\{1,2,3\} = \{1,2,3\} \qquad \{1,2,3\}\cup\{1,2,3\} = \{1,2,3\}$$

In other words, that all A_α are equal. □

Solution. 8.13 If $J \neq \varnothing, J \subseteq I$, then is it true that $\bigcup_{\alpha\in J} A_\alpha \subseteq \bigcup_{\alpha\in I} A_\alpha$? Let us break down the statement. First, suppose that $J = \{1,2,3\}$ and $I = \{1,2,3,4\}$, then the statement would be true. Intuitively, the idea is that J can be at most as large as I. If we think about the definition of big union

$$\bigcup_{\alpha\in J} A_\alpha = \{x : \exists\alpha \in I, x \in A_\alpha\},$$

so, the requirement is that at *at least one* x that is a member of A_α, then since $J \subseteq I$, it follows by definition of subsets that the equivalence holds.

Now, let's consider the second one. The definition of big intersection is $\forall\alpha \in I, x \in A_\alpha$ to be true. This however is not necessitated by the definition of subets. So it could be true, but it *does not* hold in general. □

Solution. 8.14 If $J \neq \varnothing, J \subseteq I$, then $\bigcap_{\alpha\in I} A_\alpha \subseteq \bigcap_{\alpha\in J} A_\alpha$. Now, let some arbitrary element $x \in \bigcap_{\alpha\in I} A_\alpha$, then $\forall\alpha \in I, x \in A_\alpha$. Similar to before, if $J \subseteq I$, then J is at most as big as I. Since, all $x \in A_\alpha$ for an arbitrary x, then it follows that $\forall\alpha \in J, x \in A_\alpha$. Thus, $x \in \bigcap_{\alpha\in J} A_\alpha$. □

Solutions to Exercises From Section 11

Solution. 11.1

1. This is a false statement.
2. This is a true statement.
3. This is a true statement.
4. This is not a statement.

5. This is a true statement.
6. This is a true statement.
7. This is a false statement.
8. This is a false statement.
9. This is an open sentence/formula.
10. This is a true statement.
11. This is an open sentence/formula.
12. This is a true statement. It is called a *tautology*.
13. This is a true statement.
14. This is not a statement.
15. This is technically a statement since it can be either true or false. But its truth value is unclear. It is similar to a conjecture.

□

Solutions to Exercises From Section 12

Solution. 12.1

1. P: 'The number 8 is even', Q: 'The number 8 is a power of 2', so $P \wedge Q$.
2. P: The matrix A is invertible. Then $\neg P$.
3. $x \neq y$ is the same as $\neg(x = y)$, so if $P : x = y$, then $\neg P : x \neq y$.
4. $x < y$ is the same as $\neq (x \geq y)$, so similarly to above we have $\neg P$, where P is $x \geq y$.
5. $y \geq x$ is equivalent to $\neg(y < x)$, so again we can simply state $\neg P$, where P is $y < x$.
6. P: Scheduled for Wednesday, Q: Scheduled for Friday, so $P \vee Q$.[8]
7. $P : x = 0$ and $Q : y = 0$ so, $P \wedge \neg Q$.
8. $P : x = 0$ and $Q : y = 0$, so $P \vee Q$. Similarly, to the footnote, we can transform this accordingly with predicates and quantifiers.
9. $x \in A \wedge x \notin B$.
10. $x \in A \vee x \in B$.

[8]When we later look at quantifiers and predicates, we will find that there is another more expressive way to represent the above statement: $\exists x(P(x) \vee Q(x))$ where x represents the quiz. Furthermore, we can say that there is a quiz on Wednesday or Friday but not both days: $\exists x((P(x) \vee Q(x)) \wedge \neg(P(x) \wedge Q(x)))$.

11. This one is slightly trickier. Let A be a member of the set, then $A \in \mathcal{P}(\mathbb{N})$, so now we have to 'append' the condition on A, which is $|A^{\complement}| < \infty$, so together we have $A \in \mathcal{P}(\mathbb{N}) \wedge |A^{\complement}| < \infty$.

12. H: Happy families, A: Alike, W: Unhappy in its own way. Breaking down the statement we have, if a family is happy, then it is alike, *and* if it is not happy, then it is unhappy in its own way. Therefore, $(H \Rightarrow A) \wedge (\neg H \Rightarrow W)$. If we don't want to use arrows or conditional statements, the formula looks thus: $(\neg H \vee A) \wedge (H \vee W)$.

13. H: Human beings want to be good, S: Too good, R: All the time. So, $H \wedge \neg S \wedge \neg R$.

14. P: A man should look for what is, Q: A man should look for what he think should be. Then, $P \wedge \neg Q$.

□

Solutions for Exercises from Section 13

Solution. 13.1

1. If the determinant is not zero, then the matrix is invertible.
2. If a function is differentiable, then it is continuous.
3. If a function is continuous, then it is integrable.
4. If a function is a polynomial, then it is rational.
5. If an integer is divisible by 8, then it is divisible by 4.
6. If a surface has only one side, then it is non-orientable.
7. If a series converges absolutely, then it converges.
8. If the ratio r of a geometric series is $|r| < 1$, then the series converges.
9. If a function is continuous, then the function is also integrable.
10. If the descriminant is negative, then the quadratic equation has no real solutions.
11. If you stop writing, you fail.
12. In general, if people accept facts as truth, then the facts agree with what they already believe.
13. If people agree with me, then I feel I must be wrong.

□

Solutions to Exercises from Section 14

Solution. 14.1

- A matrix A is invertible if and only if (iff) $\det(A) \neq 0$.
- A function has a constant derivative iff it is linear.
- $xy = 0$ iff $x = 0$ or $y = 0$.
- $a \in \mathbb{Q}$ iff $5a \in \mathbb{Q}$.
- An occurrence is an adventure iff one recounts it.

□

Solutions for Exercises from Section 15

Solution. 15.1 For (1):

P	Q	R	$Q \Rightarrow R$	$P \vee (Q \Rightarrow R)$
T	T	T	T	T
T	T	F	F	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	**F**
F	F	T	T	T
F	F	F	T	T

(2):

Q	R	$(Q \vee R) \Leftrightarrow (R \wedge Q)$
T	T	T
T	F	F
F	T	F
F	F	T

(3):

P	Q	$\neg(P \Rightarrow Q)$
T	T	F
T	F	T
F	T	F
F	F	F

(4):

P	Q	$\neg(P \vee Q) \vee (\neg P)$
T	T	F
T	F	F
F	T	T
F	F	T

(5):

P	Q	$(P \wedge \neg P) \vee Q$
T	T	T
T	F	F
F	T	T
F	F	F

(6):

P	Q	$(P \wedge \neg P) \wedge Q$
T	T	F
T	F	F
F	T	F
F	F	F

(7):

P	Q	$\neg(P \wedge \neg P) \Rightarrow Q$
T	T	T
T	F	T
F	T	T
F	F	T

(8):

P	Q	R	$P \vee (Q \wedge \neg R)$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	F

(9):

P	Q	$\neg(P \vee \neg Q)$
T	T	T
T	F	F
F	T	F
F	F	F

□

Solution. 15.2 First we will find it without a truth table, and then we will confirm our answer by using a truth table.
First, we need to find the global operator, in our case

$$((P \wedge Q) \vee R) \Rightarrow (R \vee S)$$

this is the '⇒'. So, we know that if we have a true hypothesis and a false conclusion, then the conditional will be false. Therefore, we require that $P = T, Q = T, R = F, S = F$. Now to check with our truth table:

P	Q	R	S	$((P \wedge Q) \vee R) \Rightarrow (R \vee S)$
T	T	T	T	T
T	T	T	F	T
T	T	F	T	T
T	T	F	F	**F!**
T	F	T	T	T
T	F	T	F	T
T	F	F	T	T
T	F	F	F	T
F	T	T	T	T
F	T	T	F	T
F	T	F	T	T
F	T	F	F	T
F	F	T	T	T
F	F	T	F	T
F	F	F	T	T
F	F	F	F	T

So, everything checks out. □

Solution. 15.3 We will follow the same approach. The formula is

$$(R \Rightarrow S) \Leftrightarrow (P \wedge Q)$$

and we are supposing that it is true. So we learned that the biconditional requires both sides to have the same truth value. Then if $R = T$ and $P, Q = T$. It follows that the biconditional is true. Let us examine the truth table:

P	Q	R	S	$((P \wedge Q) \vee R) \Rightarrow (R \vee S)$
T	T	T	T	T
T	T	T	F	F (because of S)
T	T	F	T	T
T	T	F	F	T
T	F	T	T	F (because of Q)
T	F	T	F	T
T	F	F	T	F (because of Q)
T	F	F	F	F (because of Q)
F	T	T	T	F (because of P)
F	T	T	F	T
F	T	F	T	F (because of P)
F	T	F	F	F (because of P)
F	F	T	T	F (because of P, Q)
F	F	T	F	T
F	F	F	T	F
F	F	F	F	F

□

Solutions for Exercises From Section 16

Solution. 16.1 (1):

P	Q	R	$P \wedge (Q \vee R)$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

The truth table above shows that the two expressions are logically equivalent. We could have inferred the same result by considering the distribution law.

(2):

P	Q	R	$P \vee (Q \wedge R)$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

The same reasoning from (1) holds here as well.

(3):

P	Q	$P \Rightarrow Q$	$\neg P \vee Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

In the table above I omitted the truth table for $\neg P$, if one wanted to include it that would be an extra column with the reverse truth values of P. As we can see, the truth tables match, so the expressions are logically equivalent.

(4):

P	Q	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

We can also use De Morgan's law to infer the logical equivalence of this formula.
(5):

P	Q	R	$\neg(P \vee Q \vee R)$	$\neg P \wedge \neg Q \wedge \neg R$
T	T	T	F	F
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	T	T

(6):

P	Q	R	$\neg(P \wedge Q \wedge R)$	$\neg P \vee \neg Q \vee \neg R$
T	T	T	F	F
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	T	T
F	F	T	T	T
F	F	F	T	T

(7):

P	Q	$P \Rightarrow Q$	$(P \wedge \neg Q) \Rightarrow (Q \wedge \neg Q)$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Here it's important to notice that the first and last case are sort of swapped. When $P, Q = T$, then $P \Rightarrow Q$ is clearly true but $(P \wedge \neg Q)$ is false and $Q \wedge \neg Q$ is also false, that's what gives us the desired truth value. In fact $Q \wedge \neg Q$ is always false.

(8):

P	Q	$\neg P \Leftrightarrow Q$	$(P \Rightarrow \neg Q) \wedge (\neg Q \Rightarrow P)$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	F	F

□

Solution. 16.2 I will not be making truth tables for these. However, it could be good practice to do that if you wish. Rather this solution will challenge you to slowly start thinking about proofs and try to do it in a more informal way. (1):

$$P \wedge Q = \neg(\neg P \vee \neg Q)$$

This is true. Let us use De Morgan's Law to see why. We know that by De Morgan's Law we have

$$\neg(P \wedge Q) = \neg P \vee \neg Q$$

So, if we introduce another negation to both sides we have

$$\neg\neg(P \wedge Q) = \neg(\neg P \vee \neg Q).$$

At this point we have not looked at that but we can simply remove $\neg\neg$ because it gives us the original truth value. So, the two expressions are logically equivalent.

(2): Assume that they are equivalent and we will try to simplify both sides.

$$(P \Rightarrow Q) \vee R = \neg((P \wedge \neg Q) \wedge \neg R)$$

Let us start from the LHS, we know that $P \Rightarrow Q = \neg P \vee Q$, so the expression becomes

$$\neg P \vee Q \vee R$$

This cannot be simplified further, so we move to the RHS:

$$\neg(P \wedge \neg Q \wedge \neg R) \quad \text{by Associativity}$$

Then

$$\neg P \vee Q \vee R \quad \text{by De Morgan's Law}$$

Therefore, these two expressions are also logically equivalent.

(3): Again, we assume for equivalence

$$\neg P \wedge (P \Rightarrow Q) = \neg(Q \Rightarrow P)$$

Starting from the LHS again, we have:

$$\neg P \wedge (\neg P \vee Q) = \neg P \quad \text{by Absorption}$$

We have not looked at this in the notes. Absorption is simply the idea that when we have something of the form $P \vee (P \wedge Q)$ it simplifies to P. Now, we move to the RHS:

$$\neg(Q \Rightarrow P) = \neg(\neg Q \vee P) = Q \wedge \neg P \quad \text{By Simplification and De Morgan's Law}$$

So, the final result is

$$\neg P \text{ and } Q \wedge \neg P$$

These are *not* logically equivalent.

(4):

$$\neg(P \Rightarrow Q) = P \wedge \neg Q$$

LHS:

$$\neg(\neg P \vee Q) = P \wedge \neg Q \quad \text{by De Morgan's Law}$$

Since, we can see that the LHS matches the RHS, it follows that the two expressions are logically equivalent.

(5):

$$P \vee (Q \wedge R) = (P \vee Q) \wedge R$$

LHS:

$$(P \vee Q) \wedge (P \vee R) \quad \text{by Distributive Law}$$

RHS:

$$(P \wedge R) \vee (Q \wedge R) \quad \text{by Distributive Law}$$

It follows that these are not logically equivalent.

(6):

$$P \wedge (Q \vee \neg Q) = \neg P \Rightarrow (Q \wedge \neg Q)$$

LHS simplifies to:

$$P$$

The reason is that Q and $\neg Q$ cannot both be true. RHS:

$$P \vee (Q \wedge \neg Q)$$

But notice that $Q \wedge \neg Q$ is a contradiction, this will always be false, so we simplify further to

$$P$$

These are, then, logically equivalent.

□